Chapter 35

Sodium

People say life can't exist without air, but it does under water; in fact, it started in the sea.

Richard Feynman (*a*)

Life originated in a 3.5% salt water solution (i.e., the salinity of the ocean), and humans have retained some of that heritage, since our cells are bathed in a 0.6% salt water solution. The salinity (sodium concentration) of this extracellular bath is highly regulated, because it plays an important role in determining the volume of the intracellular and extracellular fluid compartments.

This chapter begins by describing the forces that determine the distribution of total body water, and the importance of sodium in determining these forces, and then presents an organized approach to the "dysnatremias" (1,2); i.e., hypernatremia and hyponatremia.

OSMOTIC ACTIVITY

The movement of water between fluid compartments is determined by a property of fluids known as *osmotic activity*, which is a reflection of the *number of solute particles* per unit volume of solvent (3). Osmotic activity is a *colligative property*, and depends only on the number of solute particles in a fluid, and not on the electrical charge, size, or chemical behavior of the solutes. The total osmotic activity of a solution is the summed osmotic activity of all the solute particles in the solution.

Relative Osmotic Activity

When two fluid compartments are separated by a membrane that is permeable to solutes and water, the solutes in each fluid compartment will equilibrate across the membrane, and the osmotic activity will be equivalent in both fluid compartments. Since water movement follows solute movement, the volume in both fluid compartments will be equivalent. This is illustrated in the Figure 35.1 (see the panel on the left).

Effective Osmotic Activity

When two fluid compartments are separated by a membrane that is not freely permeable to solutes, the solutes will not distribute evenly in the fluid compartments, and the osmotic activity will be different in each compartment. In this situation, water moves from the fluid with the lower osmotic activity (which has a higher water content) to the fluid with the higher osmotic activity (which has a lower water content), as demonstrated in the panel on the right in Figure 35.1. The difference in osmotic activity between the fluid compartments is called the *effective osmotic activity*, and it is the force that drives the movement of water between the fluid compartments. This force is also called the *osmotic pressure*.

The relative osmotic activity between two fluid compartments is expressed as *tonicity*; i.e., if the osmotic activity in the two compartments is equal, the fluids are described as *isotonic* (see the left-side panel in Figure 35.1), and if the osmotic activity differs, the fluid with the higher osmotic activity is described as *hypertonic*, and the fluid with the lower osmotic activity is described as *hypotonic* (see the right-side panel in Figure 35.1).

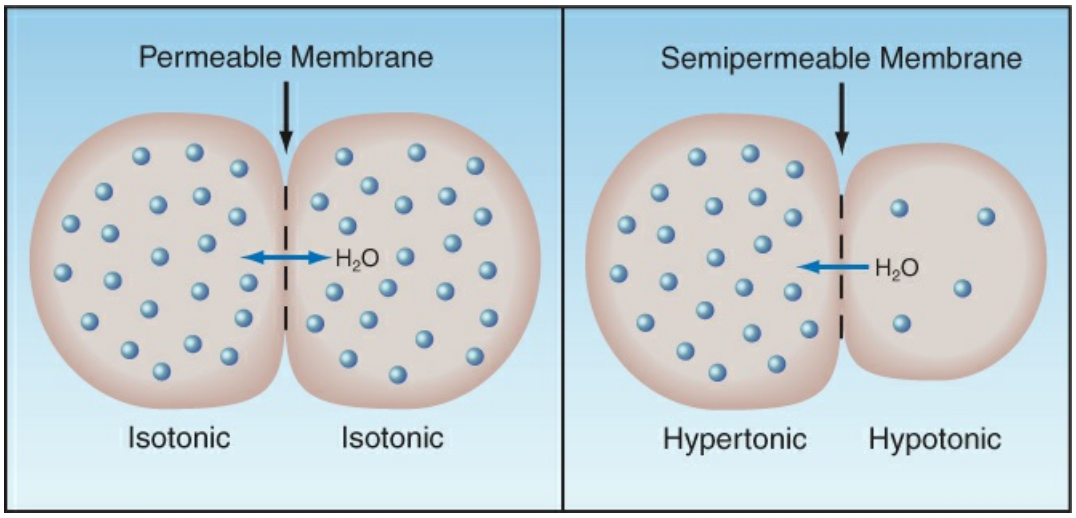


FIGURE 35.1 Illustration of the relationship between osmotic activity and water movement between fluid compartments. See text for explanation.

Summary

The following statements summarize the relationship between osmotic activity and the transcellular shift of water.

. A change in the effective osmotic activity or *tonicity* of extracellular fluid produces a transcellular water shift.

a. When the extracellular fluid is hypertonic, water moves out of cells.
b. When the extracellular fluid is hypotonic, water moves into cells.

Units of Osmotic Activity

The unit of measurement for osmotic activity is the osmole (osm), which is defined as one gram molecular weight (one mole) of a nondissociable substance, and is equivalent to 6×10^{23} particles (Avogadro's Number). Osmotic activity can be expressed in relation to the volume of water in a solution, or in relation to the volume of the solution (4,5).

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. Osmotic activity per volume of solution is called *osmolarity*, and is expressed as milliosmoles per liter (mosm/L).

. Osmotic activity per volume of water is called *osmolality*, and is expressed as milliosmoles per kilogram of H_2O (mosm/kg H_2O, or mosm/kg).

Plasma is mostly (95%) water, so the osmotic activity in plasma is typically expressed as osmolality (mosm/kg H_2O). However, there is little difference between the osmolality and osmolarity of extracellular fluid, and the two terms are often used interchangeably (5).

Conversion Factors

The following formulas (where n is the number of nondissociable particles) can be used to convert plasma solute concentrations to units of osmolality:

. For solute concentrations expressed in mEq/L:

$$\frac{\text{mEq/L}}{\text{valence}} \times \text{n} = \text{mosm/kg } H_2O \tag{35.1}$$

Thus, for univalent ions like sodium, the plasma concentration in mEq/L is equivalent to the osmotic activity in mosm/kg H_2O.

. For solute concentrations expressed in mg/dL:

$$\frac{\text{mg/dL} \times 10}{\text{mol wt}} \times \text{n} = \text{mosm/kg } H_2O \tag{35.2}$$

where mol wt is molecular weight, and the factor 10 is used to convert deciliters (dL) to liters. For example, glucose has a molecular weight of 180, so a plasma glucose concentration of 90 mg/dL is equivalent to (90 × 10/180) × 1 = 5 mosm/kg H_2O.

Plasma Osmolality

The plasma osmolality can be measured or calculated.

Measured Plasma Osmolality

The standard method for measuring plasma osmolality is the *freezing point depression* method. Solute-free water freezes at 0º C, and this temperature decreases by 1.86° C for each osmole of solute that is added to one kilogram of water. Therefore, the extent of depression of the freezing point of an aqueous solution can be used to determine the osmotic activity of the solution. This method is considered the “gold-standard” for measuring plasma osmolality, and is available in most clinical laboratories. The normal plasma osmolality is 285–295 mosm/kg H_2O.

Calculated Plasma Osmolality

Plasma osmolality can also be calculated using the concentrations of the principal solutes in plasma; i.e., sodium, chloride, glucose, and urea (3); i.e.,

$$\text{Posm} = 2 \times \text{Na} + \frac{\text{glucose}}{18} + \frac{\text{BUN}}{2.8} \quad (35.3)$$

where Posm is the plasma osmolality in mosm/kg H_2O, Na is the plasma sodium concentration in mEq/L, glucose and BUN are the plasma glucose and urea concentrations in mg/dL, and the factors 18 and 2.8 are the molecular weights of glucose and urea divided by 10, respectively (which converts their concentrations to mosm/kg H_2O). The sodium concentration is doubled to account for the negative ions (mostly chloride) that electrically balance sodium.

EXAMPLE: Using normal plasma concentrations of Na (140 mEq/L), glucose (90 mg/dL), and BUN (14 mg/dL), the plasma osmolality is: (2 × 140) + 90/18 + 14/2.8 = 290 mosm/kg H_2O. This is within the range of the measured plasma osmolality (285–295 mosm/kg H_2O).

EFFECTIVE PLASMA OSMOLALITY: Urea readily crosses cell membranes, so an increase in blood urea nitrogen (BUN) will not increase the effective osmotic activity of plasma. In other words, *azotemia is a hyperosmotic, but not a hypertonic, condition*. Therefore, the calculation of effective plasma osmolality does not include the BUN; i.e.,

$$\text{Effective Posm} = (2 \times \text{Na}) + \frac{\text{glucose}}{18} \quad (35.4)$$

EXAMPLE: Using normal plasma concentrations of Na (140 mEq/L) and glucose (90 mg/dL), the effective plasma osmolality is (2 × 140) + 90/18 = 285 mosm/kg H_2O, which is very close to the total osmolality (290 mosm/kg H_2O).

Spotlight on Sodium

If the glucose is removed from Equation 35.4, the calculated Posm is 280 mosm/kg H_2O, which is 98% of the calculated Posm that includes glucose (i.e., 285 mosm/kg H_2O). This demonstrates the following:

. Sodium is responsible for 98% of the effective plasma osmolality.

. The sodium concentration in extracellular fluid (plasma) is the principal factor that determines the distribution of total body water in the intracellular and extracellular fluid compartments, and hence it is also the principal determinant of the extracellular volume.

HYPERNATREMIA

Hypernatremia (i.e., plasma sodium >145 mEq/L) has been reported in as many as 25% of ICU patients, and *in most cases, it is acquired during the ICU stay* (6). The mechanisms and management of hypernatremia are outlined in Figure 35.2. This approach is based on the extracellular volume (ECV), and this creates three categories of hypernatremia: hypovolemic, normovolemic, and hypervolemic.

Hypovolemic Hypernatremia

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Hypernatremia associated with a low ECV is due to fluid loss that is hypotonic to plasma. The average sodium concentration in fluids that can be lost is shown in Table 35.1 (7–9). Note the following:

. All the fluids are hypotonic to plasma (i.e., have a sodium concentration <135 mEq/L), so loss of these fluids will in hypernatremia.

. All the fluids contain sodium, so loss of these fluids will result in sodium depletion as well as water depletion.

Common sources of hypotonic fluid loss in ICU patients include vomiting, diarrhea, and urine loss from diuretics or glycosuria.

Consequences

Hypernatremia increases the osmolality of the extracellular fluid, which draws water out of cells and decreases cell volumes. This effect is most prominent in the brain. However, the shrunken brain cells begin to regain volume after about 9 hours, thanks to the intracellular accumulation of "idiogenic osmolytes", and the cells can return to their normal volumes after just 48 hours (10).

TABLE 35.1 Average Sodium Concentration in Select Fluids

Normal	Na (mEq/L)	Other	Na (mEq/L)
Gastric Secretions	60	Diuretic Urine	80
Stool	25	Ileostomy Drainage	125
Urine	<10	Inflammatory Diarrhea	75
Sweat	65	Secretory Diarrhea	90

From References 7–9.

HYPERNATREMIC ENCEPHALOPATHY: The most common complication of hypernatremia is an encephalopathy that varies widely in incidence and severity. It is more likely to appear when the rise in serum sodium is rapid, and can present with lethargy, cognitive impairment, delirium, seizures and even coma (6,11). Cell shrinkage and osmotic demyelination are recognized as the culprits (11).

OTHERS: Hypernatremia has surprisingly few adverse consequences other than the encephalopathy. There are occasional reports of rhabdomyolysis (12), which often appears along with the encephalopathy (11). Hypertonicity also has negative inotropic effects, but there is no evidence that this results in heart failure. Finally, hypotonic fluid loss does not decrease the blood pressure unless fluid losses are severe, because the loss of hypotonic fluids increases the osmotic pressure in plasma, which draws fluid from the interstitial space into the plasma to help maintain intravascular volume.

Management

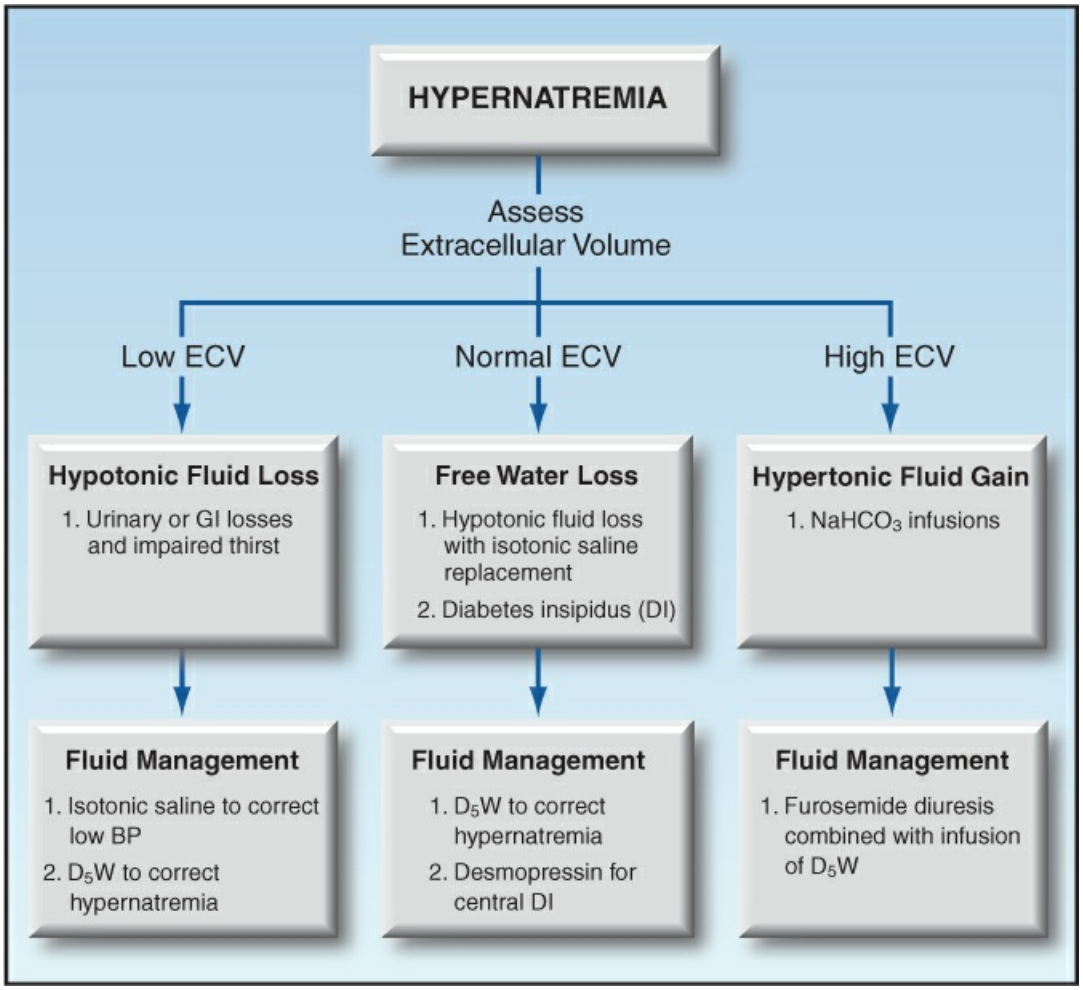
Isotonic saline is used to correct hypotension, but not to correct the hypernatremia (because it is

ineffective) (13). The recommended fluid regimens to correct the hypernatremia include the infusion of 5% dextrose in water (D_5W), or the addition of water to enteral tube feedings (e.g., 200 mL every 8 hours) (6,13). However, hypotonic fluid loss also results in the loss of sodium, so 0.45% (half-normal) saline seems more appropriate than D_5W as a replacement fluid.

RATE OF CORRECTION: Aggressive infusion of hypotonic fluids has always been discouraged because of the risk of cerebral edema, which is heightened by the accumulation of osmolytes in shrunken brain cells that helps regain the original cell volume. The optimal correction rate for hypernatremia is not known, but the traditional recommendation has been to limit the rate of correction to ≤0.5 mEq/L/hr (2,14). However, correction rates exceeding 0.5 mEq/L/hr have been used safely, without causing cerebral edema (14).

Normovolemic Hypernatremia

Hypernatremia in the setting of a normal ECV is usually the result of water loss without significant sodium loss. This condition is common in ICU patients with hypernatremia (15), and occurs when the replacement fluid is hypertonic to the fluid that is lost. (With this scenario, sodium losses are replaced, but a net water deficit remains.) Selective loss of free water is also characteristic of diabetes insipidus.



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FIGURE 35.2 Flow diagram for hypernatremia based on the extracellular volume (ECV).

Diabetes Insipidus

Diabetes insipidus (DI), is a disorder of renal water conservation, and is characterized by loss of urine that is largely devoid of solute (16,17). The underlying problem in DI is loss of the effect produced by the antidiuretic hormone (ADH), a polypeptide released by the posterior pituitary gland that promotes water reabsorption in the distal renal tubules. Three types of DI are recognized:

. *Central DI* is characterized by failure of ADH release from the posterior pituitary (16). Common causes include surgical resection of a pituitary tumor, traumatic brain injury, subarachnoid hemorrhage, and autoimmune diseases (16).

. *Nephrogenic DI* is characterized by impaired renal responsiveness to ADH. Possible causes in critically ill patients include drugs (amphotericin, aminoglycosides, dopamine, lithium, etc.), hypokalemia, and the recovery phase of ATN (17).

. *Gestational DI* is the result of a vasopressinase enzyme that is produced by the placenta and inactivates arginine vasopressin (ADH) (18). Production of the enzyme increases throughout pregnancy, and the DI usually appears at the end of the second, or in the third, trimester.

DIAGNOSIS: The hallmark of DI is a dilute urine in the face of hypertonic plasma. In central DI, the urine osmolarity is often below 200 mosm/L, whereas in nephrogenic DI, the urine osmolarity is 200–500 mosm/L (19). The diagnosis of DI is based on the response of the urine osmolality to fluid restriction. Failure of the urine osmolarity to increase more than 30 mosm/L in the first few hours of complete fluid restriction will identify a case of DI. The response to vasopressin can then differentiate central from nephrogenic DI; i.e., in central DI, the urine osmolality increases by at least 50% after vasopressin administration, whereas in nephrogenic DI, the urine osmolality is unchanged after vasopressin.

DESMOPRESSIN: In central DI, renal water retention can be restored with the administration of *desmopressin*, a synthetic analog of vasopressin that lasts longer, and has no vasoconstrictor effects. Desmopressin is available as an oral tablet, an intranasal spray, and a parenteral preparation for subcutaneous or intravenous injection. Only 5% of the drug is absorbed from the GI tract (20), so the oral route is least desirable. The parenteral dose of desmopressin is 1 μg by subcutaneous injection every 12 hours, or 2 μg by IV injection every 12 hours (20).

NSAIDS: The approach to nephrogenic DI involves removing any offending drug, but this does not always correct the problem. When this occurs, nonsteroidal anti-inflammatory drugs (NSAIDs) can increase renal responsiveness to ADH (because prostaglandins block the actions of ADH, and NSAIDs inhibit prostaglandin synthesis). Indomethacin (2 mg/kg/day) has been more successful than other NSAIDs in restoring ADH responsiveness (21).

Hypervolemic Hypernatremia

Hypernatremia with a high extracellular volume can be the result of sodium bicarbonate infusions, but it is more frequently associated with excessive infusions of isotonic saline for

hypotonic fluid loss.

Management

In patients with normal renal function, excess sodium and water are excreted rapidly. When renal sodium excretion is impaired, it might be necessary to increase renal sodium excretion with a diuretic (e.g., furosemide). Because the urinary sodium concentration during furosemide diuresis (~80 mEq/L) is less than the plasma sodium concentration, diuresis can aggravate the hypernatremia. Therefore, the diuresis should be combined with infusions of D_5W to correct the hypernatremia.

HYPONATREMIA

Hyponatremia (plasma [Na] < 135 mEq/L) is considered the most common electrolyte abnormality in clinical practice (22), and is found in 30–40% of hospitalized patients (23).

Pseudohyponatremia

Plasma has an aqueous phase and a solid (nonaqueous) phase. Small electrolytes like sodium are confined to the aqueous phase, while larger molecules like lipids and proteins are located in the solid phase. The automated measurements performed in the clinical laboratory include both phases of plasma (24), and thus the measured plasma sodium could underestimate the actual (aqueous phase) sodium level. However, 95% of the plasma is water (i.e., aqueous phase), so there is normally little difference between the measured and actual sodium concentrations.

However, a problem arises when there is a marked increase in protein or lipid levels in blood. In this situation, there is an increase in the nonaqueous or solid phase of plasma, which will decrease the measured plasma sodium concentration, while the actual (aqueous phase) sodium concentration is unchanged. This spurious decrease in the measured plasma sodium is called *pseudohyponatremia* (24), and it becomes significant when plasma protein levels reach 12 g/dL (normal range is 5.5–8 g/dL), and triglyceride levels rise to 1,500 mg/dL (normal range is 25–175 mg/dL). The clinical conditions that can produce pseudohyponatremia are listed in Table 35.2 (24).

TABLE 35.2 Clinical Conditions Associated with Pseudohyponatremia

Mechanism	Conditions
Hyperproteinemia	HIV Disease Immunoglobulin Infusions Monoclonal Gammopathies Multiple Myeloma Waldenström's Macroglobulinemia
Hypertriglyceridemia	Alcohol Abuse Diabetic Ketoacidosis Hyperlipidemias (types I and V) Pancreatitis Type 2 Diabetes (poorly controlled)

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Hypercholesterolemia	Biliary Obstruction Hepatitis Intrahepatic Cholestasis Primary Biliary Cirrhosis

From Reference 24.

Diagnosis

Pseudohyponatremia can be uncovered by measuring the plasma osmolality, which will be normal (285–295 mosm/kg H_2O) with pseudohyponatremia, and will be reduced (<275 mosm/kg H_2O) with true hyponatremia.

Hyperglycemia

Hyperglycemia is another potential cause of spurious hyponatremia. When glucose does not readily enter cells, the resultant hyperglycemia draws fluid out of cells. This increases the aqueous phase of plasma, and has a dilutional effect on sodium. For every 100 mg/dL increase in blood glucose, there is a 2.4 mEq/L decrease in the serum sodium concentration (22,25). This is expressed in the following equation:

$$\text{Corrected [Na]} = \text{Measured [Na]} + \left[2.4 \times \frac{\text{glucose} - 100\ (\text{mg/dL})}{100\ (\text{mg/dL})} \right] \quad (35.5)$$

EXAMPLE: If a patient with a serum sodium concentration of 130 mEq/L also has a blood glucose of 600 mg/dL, the corrected serum sodium is 130 + (2.4 × 5) = 142 mEq/L.

The Approach

The evaluation and management of true hyponatremia can be organized around the extracellular volume (ECV), as shown in Figure 35.3. This approach is similar to the one used for hypernatremia, and it has three types of hyponatremia: hypovolemic, normovolemic, and hypervolemic.

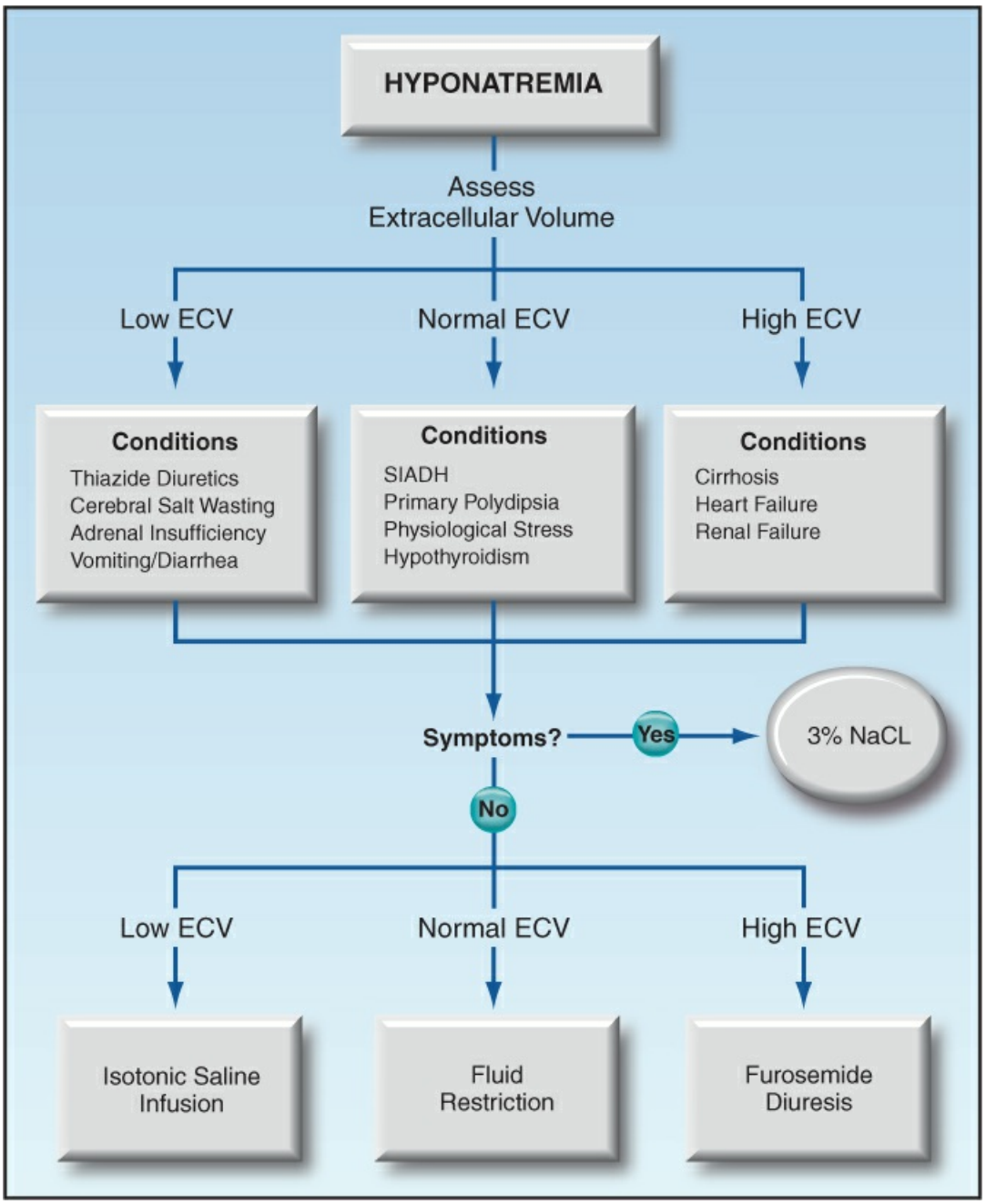


FIGURE 35.3 Flow diagram for hyponatremia based on the extracellular volume (ECV).

Hypovolemic Hyponatremia

Hyponatremia with a low ECV is the result of sodium loss in the urine or GI tract, combined with impaired water excretion by the kidneys (26). Water retention is needed to produce the hyponatremia (because loss of hypotonic fluids results in *hyper*natremia), and is the result of the following events: (*a*) the sodium losses result in hypovolemia, which stimulates the renin-angiotensin-aldosterone system, (*b*) the released angiotensin stimulates the release of antidiuretic hormone (ADH) from the posterior pituitary gland, and (*c*) ADH promotes water retention by the kidneys.

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Etiologies

The clinical conditions that promote hypovolemic hyponatremia are listed in Figure 35.3. The most cited causes are thiazide diuretics, adrenal insufficiency, and cerebral salt wasting.

THIAZIDE DIURETICS: Hypernatremia is reported in about one-third of patients being treated with a thiazide diuretic, and usually appears after 3 weeks of treatment (27). The mechanism for the hyponatremia is complex, and beyond the intent of this chapter.

ADRENAL INSUFFICIENCY: Primary adrenal insufficiency involves mineralocorticoid deficiency, which promotes renal sodium wasting. The hyponatremia usually appears in the chronic (not acute) form of illness. Secondary adrenal insufficiency (which originates in the hypothalamus or pituitary) is primarily a deficiency of glucocorticoids, and is not associated with hyponatremia.

CEREBRAL SALT WASTING: Cerebral salt wasting is a condition of urinary sodium loss and hyponatremia that is associated with neurologic disorders, especially subarachnoid hemorrhage, traumatic brain injury, CNS infections, Guillain-Barre Syndrome, and neurosurgery (28).

Diagnostic Considerations

The source of sodium loss (i.e., kidneys or GI tract) is usually apparent, but if not, a spot urine sodium can help to distinguish renal from extrarenal losses; i.e., urine $Na^+ > 20$ mEq/L suggests a renal source of sodium loss, while a urine $Na^+ < 20$ mEq/L suggests an extrarenal site of sodium loss.

Normovolemic Hyponatremia

Hyponatremia with a normal ECV is the result of water retention with minimal sodium loss. This is usually the result of uncontrolled, “nonosmotic” release of ADH from the pituitary, or excessive water intake.

Etiologies

This causes of normovolemic hyponatremia include: the syndrome of inappropriate ADH (SIADH), primary polydipsia, physiological stress, and (rarely) hypothyroidism (see Figure 35.3).

SIADH: SIADH is a condition of uncontrolled or nonosmotic release of ADH that is associated with a variety of other conditions, including neoplasms, infections, and drugs. The hallmark of SIADH is the combination of hypotonic plasma, inappropriately concentrated urine (urine osmolality >100 mosm/kg H_2O) and an elevated urine sodium (>20 mEq/L) (29). This condition can be difficult to distinguish from cerebral salt wasting, and the only difference may be the ECV (which is decreased with cerebral salt wasting, and normal with SIADH).

PHYSIOLOGICAL STRESS: Physiological stress is well known for promoting the release of ACTH from the anterior pituitary, but a lesser known component of the stress response is the release of ADH from the posterior pituitary. (Teleologically, water retention from the actions of ADH should be advantageous during periods of stress, when insensible losses are likely to be high.)

Stress-induced release of ADH would explain the increased risk of hyponatremia in postoperative patients (22), and this should also apply to critically ill patients. The popularity of hypotonic IV fluids like Ringer's lactate adds further to the risk of hyponatremia in hospitalized patients (30). (See Table 10.2 for the osmolarity of intravenous fluids.)

PRIMARY POLYDIPSIA: Water intoxication (primary polydipsia) is a source of hyponatremia that occurs primarily in patients with developmental or psychiatric disorders (31). Patients with hyponatremia from polydipsia also have impaired free water clearance by the kidneys (32), and both factors may be necessary to produce sustained hyponatremia.

Another example of hyponatremia caused by heavy consumption of a dilute fluid is *beer potomania* (33), which is typically found in chronic alcoholics who are poorly nourished and are prone to binge drinking.

HYPOTHYROIDISM: Although hypothyroidism is considered a cause of hyponatremia, the evidence for this is not convincing. For example, a retrospective analysis of 8,000 patients with hypothyroidism revealed that 5% of the patients had hyponatremia, but almost all (98%) of these patients had other conditions that could have caused the hyponatremia (34). Despite the difficulty establishing a causal link between hypothyroidism and hyponatremia, a serum TSH assay is recommended in the evaluation of hyponatremia of undetermined etiology (34).

Diagnostic Considerations

When the cause of normovolemic hyponatremia is unclear, the urine osmolality can help to distinguish between SIADH and primary polydipsia: i.e., in SIADH, the unregulated actions of ADH produce a concentrated urine (urine osmolality 300–500 mosm/kg H_2O), whereas water intoxication produces a dilute urine (i.e., urine osmolality <200 mosm/kg H_2O).

Hypervolemic Hyponatremia

Hypervolemic hyponatremia is the result of salt and water retention, with water retention exceeding sodium retention. This condition occurs in advanced heart failure, cirrhosis, and renal failure. Renal failure is associated with a high urine [Na^+] (>20 mEq/L), while the urine [Na^+] is low (<20 mEq/L) in heart failure and cirrhosis (except when a diuretic is active).

MANAGEMENT OF HYPONATREMIA

The management of hyponatremia is determined by two factors: the ECV, and the presence or absence of hyponatremic encephalopathy.

Hyponatremic Encephalopathy

Hyponatremia results in a hypotonic extracellular fluid, which then causes fluid to move across the blood-brain barrier and into brain parenchyma. This process culminates in diffuse cerebral edema with an increased intracranial pressure (35), and when this condition produces symptoms, it is called *hyponatremic encephalopathy* (36). This is considered an immediate threat to life, and thus requires prompt action to reduce the cerebral edema with hypertonic saline.

Hyponatremic encephalopathy is more likely appear when the hyponatremia is acute (i.e., develops in less than 48 hours), but there is no correlation between the severity of the hyponatremia and the appearance or severity of the encephalopathy (36). As a result, *the plasma Na^+ concentration has no role in the decision about initiating hypertonic saline*.

Symptoms

Hyponatremic encephalopathy can present with a wide variety of symptoms, but the earliest symptoms usually include nausea, vomiting, headache, and ataxia, while advanced symptoms can include agitation, confusion, seizures, and coma. Although the early symptoms can be nonspecific, *the presence of any symptom that suggests possible cerebral edema is an indication for rapid correction of the hyponatremia with hypertonic saline*.

Hypertonic (3%) Saline

The treatment of symptomatic hyponatremia with hypertonic saline is summarized in Table 35.3. A 3% NaCL solution (Na^+ = 513 mEq/L, osmolarity = 1,026 mosm/L) is given in bolus doses of 2 mL/kg (or 150 mL), which are repeated every few hours until the symptoms resolve. Each bolus dose can increase the sodium concentration by as much as 2 mEq/L, and an increase of 5–6 mEq/L should produce some improvement in symptoms. If there is no improvement at this point, it is possible that hyponatremia is not the cause of the symptoms.

The 3% NaCL solution can be delivered safely through a peripheral vein (36,37), and treatment continues until the symptoms resolve. However, there are limits placed on the rate and extent of Na correction, as explained next.

TABLE 35.3 Hypertonic Saline for Symptomatic Hyponatremia

1. Use 3% NaCL (Na^+ = 153 mEq/L, osmolarity = 1,026 mosm/L).
2. Give IV bolus doses of 2 mL/kg (or 150 mL), which are repeated every few hours until the symptoms resolve. This regimen can be given via peripheral veins.
3. Assess after the serum Na^+ has increased by 5 mEq/L – if there is no improvement, consider another cause for the symptoms.
4. The following limits are recommended as a preventive measure for osmotic demyelination.
   a. The initial increment in serum Na^+ should not exceed 5 mEq/L in 2 hrs, or 10 mEq/L in 5 hrs.
   b. The daily increment in serum Na^+ should not exceed 10 mEq/L on the first day, and 8 mEq/L on subsequent days.
   c. The final plasma Na^+ should not exceed 130 mEq/L.

From References 22 and 36.

Osmotic Demyelination Syndrome

Correction of the sodium that is too rapid, or overshoots the normal sodium concentration can produce a devastating neurologic disorder caused by osmotic damage to the myelin sheath of brainstem neurons. This *osmotic demyelinating syndrome*, also known as ”central pontine myelinolysis”, has a biphasic course. The initial presentation includes varying degrees of altered mentation, along with seizures, but this improves, only to be followed at a later time by profound deterioration in overall health, with dysarthria, oculomotor dysfunction, quadriparesis, and even

locked-in syndrome. (38). Chronic hyponatremia poses a greater risk for this complication than acute (within 48 hrs) hyponatremia.

To reduce the risk of osmotic demyelination, the maximum rate of correction for Na^+ is limited to 8–10 mEq/L over any 24-hour period, and the end-point of Na^+ correction is a serum concentration of 130 mEq/L. (22,36). This end-point is below the normal range, but it provides a buffer against overshooting the end-point.

Overcorrection

Overcorrection of the serum sodium occurs in as many as 25% of cases (39), even when preventive measures are used. The tendency for overcorrection can be attributed to a spontaneous water diuresis that occurs in some patients during treatment with hypertonic saline. This diuresis decreases the extracellular volume, which concentrates the sodium and augments its rate of correction.

The water diuresis is more frequent in hyponatremia caused by drugs (especially thiazides) and by water intoxication. This diuresis can be prevented or terminated by desmopressin.

DESMOPRESSIN: Desmopressin (a synthetic analog of vasopressin) can be used as shown in Table 35.4 (40,41). It should be started when there is an abrupt increase in urine flow (indicating water diuresis) during corrective therapy with hypertonic saline, and it can be used routinely in cases of hyponatremia that have a high risk of overcorrection (i.e., hyponatremia from adrenal insufficiency, drugs, and primary polydipsia.) Fluid restriction is mandatory during therapy with desmopressin, to prevent the development of severe hyponatremia.

TABLE 35.4 Desmopressin to Prevent or Reverse Overcorrection of Hyponatremia

Indications: • If the increment in serum Na^+ exceeds the limits listed in Table 35.3. • If there is an abrupt increase in urine output (e.g., >100 mL/hr) during corrective therapy with hypertonic saline • As a routine measure in conditions where overcorrection is a risk (see text).
Dose: • 2 µg by subcutaneous injection every 12 hrs.
Caveats: • Desmopressin is not advised when the hyponatremia is due to unregulated release of ADH. • Fluid restriction is mandatory during R_x with desmopressin.

Asymptomatic Hyponatremia

Rapid correction of the hyponatremia is not necessary in asymptomatic patients. Instead, the fluid management is based on the ECV, as shown in Figure 35.3.

Hypovolemia

In cases of hypovolemic hyponatremia, isotonic saline will help to improve blood pressure and tissue perfusion, but the hyponatremia may not show significant improvement until the

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responsible conditions are treated or eliminated.

Normovolemia

The problem in normovolemic hyponatremia is water excess without sodium excess, so fluid restriction is indicated. The usual goal is a daily fluid intake that is 500 mL less than the daily urine output (26). Fluid restriction is notoriously intolerable for patients, but the following drug options available for patients with SIADH.

DEMECLOCYCLINE: Demeclocycline is a tetracycline derivative that blocks the effects of ADH in the renal tubules. It is intended for patients with chronic SIADH who do not tolerate fluid restriction, and is given in a dose of 600–1,200 mg daily in divided doses (26). Maximum effect takes several days, and success is variable. Demeclocycline can be nephrotoxic, so monitoring renal function is advised while using the drug.

THE VAPTANS: The vaptans are vasopressin antagonists that block the actions of vasopressin (ADH) in the kidneys and elsewhere. There are two drugs in this class: conivaptan and tolvaptan. Conivaptan can be given intravenously, while tolvaptan is an oral medication. Both agents can increase the serum sodium by 6 or 7 mEq/L (42,43). These drugs are expensive and are not currently popular.

Hypervolemia

Patients with hypervolemic hyponatremia are both salt and water overloaded, with the water overload exceeding the salt overload. Furosemide diuresis could theoretically correct this problem. However, continued activation of the renin-angiotensin-aldosterone system in heart failure and cirrhosis helps to maintain the hyponatremia.

A FINAL WORD

What You See is Not What You Get

The fundamental truth about hypernatremia and hyponatremia is that they are disorders of water balance, not sodium balance. This “sleight of hand” is unique among electrolyte disorders; e.g., hypomagnesemia is a sign of magnesium depletion, but hyponatremia is a sign of water overload (absolute or relative), not sodium depletion. Understanding this basic distinction may help you to navigate through the maze of possibilities when confronting a patient with hypernatremia or hyponatremia.

References

a. Feynman R. *The Unscientific Age*. Lecture at the University of Washington, 1963.

General Reviews

1. Seay NW, Leheich RW, Greenberg A. Diagnosis and management of disorders of body tonicity— hyponatremia and hypernatremia: core curriculum 2020. Am J Kidney Dis 2020; 75:272–286.
2. Sterns RH. Disorders of plasma sodium—causes, consequences, and correction. N Engl J Med 2015; 372:5565.

Osmotic Activity

3. Rose BD, Post TW. The total body water and the plasma sodium concentration. In: *Clinical Physiology of Acid-Base and Electrolyte Disorders*. 5th ed. New York, NY: McGraw-Hill, 2001; 241–257.
4. Gennari FJ. Current concepts. Serum osmolality. Uses and limitations. N Engl J Med 1984; 310:102–105.
5. Erstad BL. Osmolality and osmolarity: narrowing the terminology gap. Pharmacother 2003; 23:1085–1086.

Hypernatremia

6. Chand R, Chand R, Goldfarb D. Hypernatremia in the intensive care unit. Curr Opin Nephrol Hypertens 2022; 31:199–204.
7. Gennari FJ, Weise WJ. Acid-base disturbances in gastrointestinal disease. Clin J Am Soc Nephrol 2008; 3:1861–1868.
8. Bates GP, Miller VS. Sweat rate and sodium loss during work in the heat. J Occup Med Toxicol 2008; 3:4 (open access article).
9. Stason WB, Cannon PJ, Heinemann HO, Laragh JH. Furosemide: A clinical evaluation of its diuretic action. Circulation 1966; 34:910–920.
10. Verbalis JG. Brain volume regulation in response to changes in osmolality. Neuroscience 2010; 168:862–870.
11. Naik KR, Saroja AO. Seasonal postpartum hypernatremic encephalopathy with osmotic extra-pontine myelinolysis and rhabdomyolysis. J Neurol Sci 2010; 291:5–11.
12. Denman JP. Hypernatremia and rhabdomyolysis. Med J Aust 2007; 187:527–528.
13. Al-Absi A, Gosmanova EO, Wall BM. A clinical approach to the treatment of chronic hypernatremia. Am J Kidney Dis 2012; 60:1032–1038.
14. Chauhan K, Pattharanitima P, Patel P, et al. Rate of correction of hypernatremia and health outcomes in critically ill patients. Clin J Am Soc Nephrol 2019; 14:656–663.
15. Pokaharel M, Block CA. Dysnatremia in the ICU. Curr Opin Crit Care 2011; 17:581–593.
16. Tomkins M, Lawless S, Martin-Grace J, et al. Diagnosis and management of central diabetes insipidus in adults. J Clin Endocrinol Met 2022; 107:2701–2715.
17. Garofeanu CG, Weir M, Rosas-Arellano MP, et al. Causes of reversible nephrogenic diabetes insipidus: a systematic review. Am J Kidney Dis 2005; 45:626–637.
18. Gestational diabetes insipidus: Diagnosis and management. Best Pract Res Clin Endocrinol Met 2020; 34:101384.
19. Geheb MA. Clinical approach to the hyperosmolar patient. Crit Care Clin 1987; 3:797–815
20. Bichet DG. Arginine vasopressin deficiency (central diabetes insipidus): Treatment. www.UpToDate.com (Accessed April 4, 2024)
21. Libber S, Harrison H, Spector D. Treatment of nephrogenic diabetes insipidus with prostaglandin synthesis inhibitors. J Pediatr 1986; 108:305–311.

Hyponatremia

22. Spasovski G, Vanholder R, Allolio B, et al. Clinical practice guideline on diagnosis and management of hyponatremia. Intensive Care Med 2014; 40:320–331.
23. Androgué HJ, Tucker BM, Madias NE. Diagnosis and management of hyponatremia: A review. JAMA 2022; 328:289–291.
24. Aziz F, Sam R, Lew SQ, et al. Pseudohyponatremia: Mechanism, diagnosis, clinical associations and management. J Clin Med 2023; 12:4076.
25. Hiller TA, Abbott RD, Barrett EJ. Hyponatremia: evaluating the correction factor for hyperglycemia. Am J Med 1999; 106:399–403.
26. Verbalis JG, Goldsmith SR, Greenberg A, et al. Hyponatremia treatment guidelines 2007: Expert panel recommendations. Am J Med 2007; 120(Suppl):S1–S21.
27. Klhůfek J, Šálek T. Thiazide-associated hyponatremia in internal medicine patients: analysis of epidemiological and biochemical profiles. Postgrad Med 2022; 134:487–493.
28. Cui H, He G, Yang S, et al. Inappropriate antidiuretic hormone secretion and cerebral salt-wasting syndromes in neurological patients, Front Neurosci 2019; 13:1170.
29. Martin-Grace J, Tomkins M, O'Reilly MW, et al. Approach to the patient: hyponatremia and the syndrome of inappropriate antidiuresis (SIAD). J Clin Endocrinol Metab 2022; 107:2362–2376.
30. Moritz ML, Ayus JC. Maintenance intravenous fluids in acutely ill patients. N Eng J Med 2015; 373:1350–1360.
31. Ahmad L, Goldman MB. Primary polydipsia: update. Best Pract Res Clin Endocrinol Metab 2020; 34:101469.
32. Goldman MB, Luchins DL, Robertson GL. Mechanisms of altered water metabolism in psychotic patients with polydipsia and hyponatremia. N Engl J Med 1988; 318:397–403.
33. Sanghvi SR, Kellerman PS, Nanovic L. Beer potomania: an unusual cause of hyponatremia at high risk of complications from rapid correction. Am J Kidney Dis 2007; 50:673–680.
34. Wolf P, Beiglblöck H, Smaijs S, et al. Hypothyroidism and hyponatremia: rather coincidence than causality. Thyroid 2017; 27:611–615.
35. Arieff AI, Ayus JC. The pathogenesis of hyponatremic encephalopathy. Current concepts. Chest 1993; 103:607–610.
36. Achinger SG, Ayus JC. Treatment of hyponatremic encephalopathy in the critically ill. Crit Care Med 2017; 45:1762–1771.
37. Metheny NA, Moritz ML. Administration of 3% sodium chloride via a peripheral vein. J Infus Nurs 2021; 44:94–102.
38. Singh TD, Fugate JE, Rabinstein AA. Central pontine and extrapontine myelinolysis: a systematic review. Eur J Neurol 2014;

A L G r a w a n y

21:1443–1450.
39. Baek SH, Jo YH, Ahn S, et al. Risk of overcorrection in rapid intermittent bolus vs slow continuous infusion therapies of hypertonic saline for patients with symptomatic hyponatremia. The SALSA randomized clinical trial. JAMA Intern Med 2021; 181:81–92.
40. Achinger SG, Ayus JC. Use of desmopressin in hyponatremia: foe and friend. Kidney Med 2019; 1:65–70.
41. Sood L, Sterns RH, Hix JK, et al. Hypertonic saline and desmopressin: a simple strategy for safe correction of severe hyponatremia. Am J Kidney Dis 2013; 61:571–578.
42. Lehrich RW, Greenberg A. Hyponatremia and the use of vasopressin receptor antagonists in critically ill patients. J Intensive Care Med 2012; 27:207–218.
43. Zeltser D, Rosansky S, van Rensburg H, et al. Assessment of efficacy and safety of intravenous conivaptan in euvolemic and hypervolemic hyponatremia. Am J Nephrol 2007; 27:447–457.

Chapter 36

Potassium

No knowledge can be more satisfactory to a man than that of his own frame, its parts, their functions and actions.

Thomas Jefferson (*a*)

The last chapter introduced the notion that the ocean served as extracellular fluid for the first unicellular organisms. Seawater is essentially a saline solution that is rich in sodium and chloride, and "poor" in potassium, similar to our extracellular fluid. Why there is so little potassium in seawater is not entirely explained, but the paucity of potassium in our extracellular fluid is the consequence of a membrane pump that keeps potassium in cells. About 98% of the total body potassium is located inside cells, and only 2% remains in the extracellular fluid (1–3). As a result, monitoring the plasma (extracellular) potassium level as an index of total body potassium is like evaluating an iceberg by its tip. With this limitation in mind, this chapter describes the causes and consequences of abnormalities in the plasma potassium concentration.

POTASSIUM BASICS

Potassium Distribution

The intracellular preponderance of potassium is the result of a sodium–potassium (Na^+–K^+) exchange pump on cell membranes that moves Na^+ out of cells and moves K^+ into cells in a 3:2 ratio (1). One of the major roles of this pump is to create a voltage gradient across cell membranes in "excitable" tissues (i.e., nerves and muscle), which allows for the transmission of electrical impulses in these tissues.

The marked difference between intracellular and extracellular K^+ is illustrated in Figure 36.1. The total body potassium in healthy adults is about 50–55 mEq per kg body weight (1). Using the conservative estimate of 50 mg/kg in a 70 kg adult yields a total body potassium of 3,500 mEq, with 70 mEq (2%) in extracellular fluid. Because the plasma accounts for about 20% of the extracellular fluid, the *potassium content of plasma* will be about 15 mEq, which *is only 0.4% of the total body potassium*. This emphasizes the limited size of the K^+ pool that is available for evaluating total body K^+.

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Serum Potassium

The relationship between total body K^+ and serum (plasma) K^+ is shown in Figure 36.2 (4,5). Note the curvilinear shape of the curve, with the flat portion of the curve in the region of potassium deficiency. In an average-sized adult with a normal serum K^+ of 4 mEq/L, a total body K^+ deficit of 200–400 mEq is required to produce a decrease in plasma K^+ of 1 mEq/L, while a total body K^+ excess of 100–200 mEq is required to produce a similar (1 mEq/L) increase in plasma K^+ (5). Therefore, *for a given change in serum K^+, the change in total body K^+ is twofold greater with K^+ depletion (hypokalemia) than with K^+ excess (hyperkalemia)*. The larger K^+ deficit associated with hypokalemia is due to the large pool of intracellular K^+ that can replenish extracellular K^+ (and help to maintain serum K^+) when K^+ is lost.

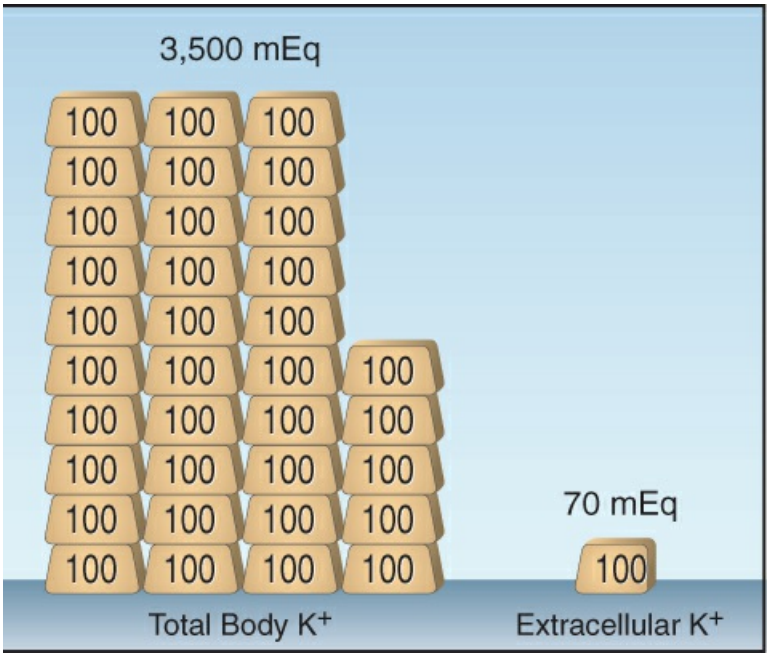


FIGURE 36.1 IIllustration of the marked difference between intracellular and extracellular potassium in a 70 kg adult with an estimated total body potassium of 50 mEq/kg. Each gold bar represents 100 mEq of potassium.

Potassium Excretion

Small amounts of K^+ are lost in stool (5–10 mEq/day) and sweat (0–10 mEq/day), but the majority of K^+ loss is in urine (40–120 mEq/day, depending on K^+ intake) (1).

Renal Excretion

Most of the K^+ that is filtered at the glomerulus is passively reabsorbed in the proximal tubules (along with sodium and water), and K^+ is then secreted in the distal tubules and collecting ducts (1). Potassium excretion in urine is primarily a function of K^+ secretion in the distal nephron, which is controlled by plasma K^+ and (primarily by) aldosterone. When renal function is normal, the capacity for renal K^+ excretion is great enough to prevent a sustained rise in serum K^+ in response to an increased K^+ load (1).

ALDOSTERONE: Aldosterone is a mineralocorticoid that is released by the adrenal cortex in

response to an increase in plasma K^+ (and angiotensin II), and it stimulates K^+ secretion in the distal nephron (and hence increases urinary K^+ excretion). Potassium secretion in the distal nephron is linked to sodium reabsorption, so aldosterone also promotes sodium and water retention. The diuretic *spironolactone* acts by blocking the actions of aldosterone in the kidneys. As a result, spironolactone is a *potassium-sparing diuretic*.

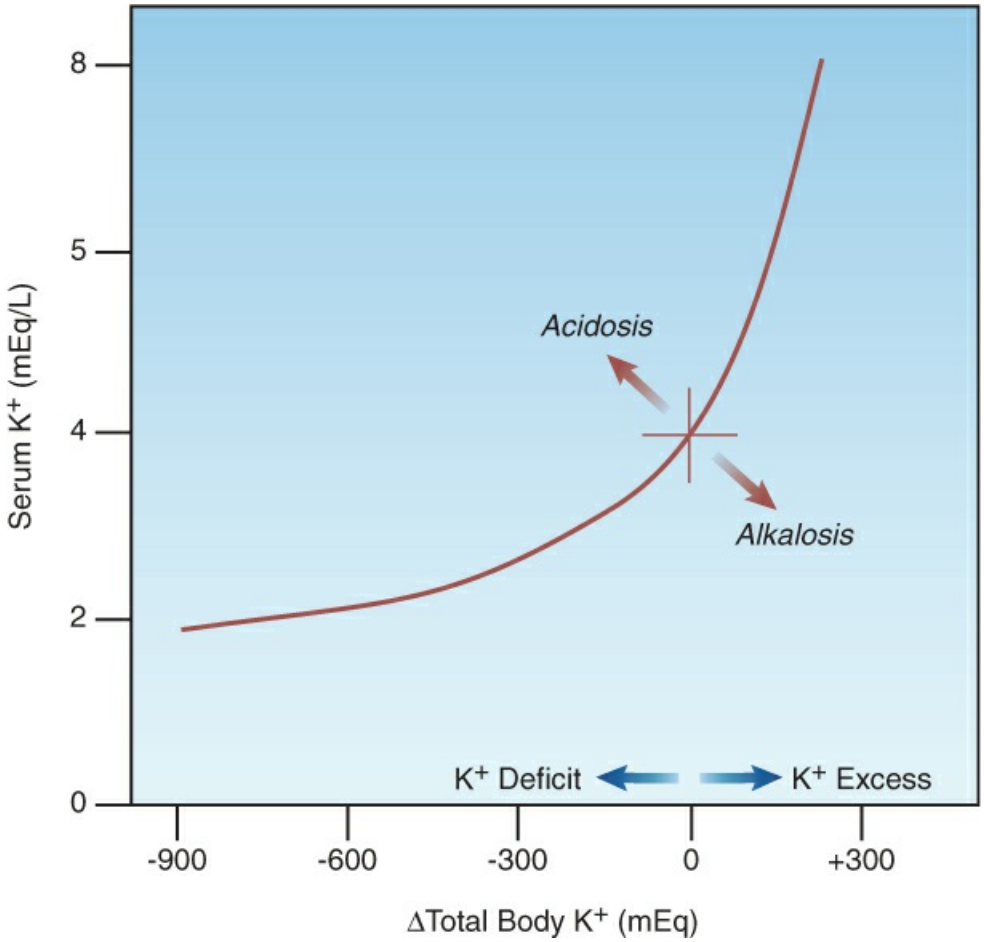


FIGURE 36.2 Relationship between the serum potassium concentration and total body potassium. Redrawn from Reference 4.

HYPOKALEMIA

Hypokalemia (serum $K^+ < 3.5$ mEq/L) can be the result of K^+ movement into cells (transcellular shift), or K^+ depletion from renal and extrarenal losses (3–6).

Transcellular Shift

The movement of K^+ into cells is facilitated by β_2-adrenergic receptors on cell membranes in muscle, and this explains the decrease in serum K^+ associated with *inhaled β_2-agonist bronchodilators* (e.g., albuterol) (7). This effect is mild (≤0.5 mEq/L) in the usual therapeutic doses (7), but is more significant when inhaled β_2-agonists are used in combination with diuretics

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(8). Other factors that promote K^+ movement into cells include *alkalosis* (respiratory or metabolic), *hypothermia* (accidental or induced), and *insulin*. Alkalosis has a variable and unpredictable effect on serum K^+ (9). Hypothermia causes a transient drop in serum K^+ that resolves on rewarming (10).

Potassium Depletion

Potassium depletion can be the result of K^+ loss via the kidneys or gastrointestinal tract. The site of K^+ loss (renal or extrarenal) is usually obvious, but can be identified by measuring spot urine potassium and chloride concentrations, as shown in Figure 36.3.

Renal Potassium Loss

The leading cause of renal K^+ loss is *diuretic therapy*. Other causes likely to be seen in the ICU include nasogastric drainage, alkalosis, and magnesium depletion. Nasogastric drainage has a low concentration of K^+ (10–15 mEq/L), but the resulting loss of volume and H^+ promotes K^+ loss in the urine. The urine chloride is low (<15 mEq/L) with nasogastric drainage and alkalosis, and it is high (>25 mEq/L) with diuretic therapy and magnesium depletion. *Magnesium depletion* impairs K^+ reabsorption in the renal tubules and *may play a very important role in promoting K^+ depletion* in critically ill patients, particularly those receiving diuretics (11).

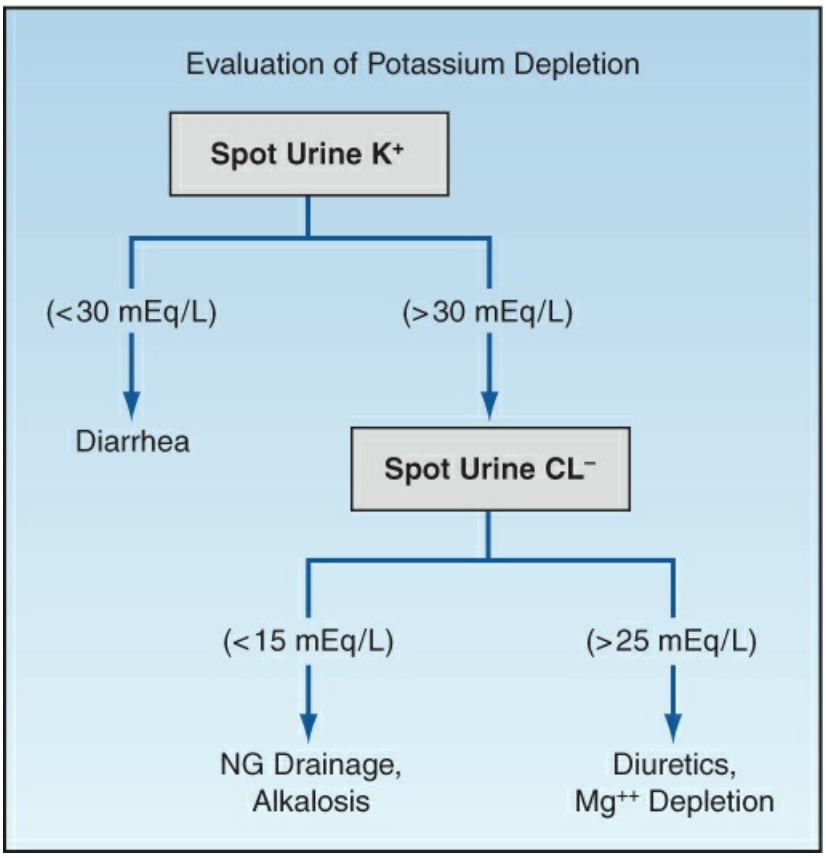


FIGURE 36.3 Evaluation to identify the site of potassium loss.

Extrarenal Potassium Loss

The major cause of extrarenal K^+ loss is *diarrhea*. Normal K^+ loss in stool is only 5–10 mEq daily. In secretory and inflammatory diarrhea, the concentration of K^+ in stool is 15–40 mEq/L, and the daily stool volume can reach 10 liters in severe cases. Therefore, K^+ losses can reach 400 mEq daily in severe cases of inflammatory or secretory diarrhea (12).

Clinical Manifestations

Severe hypokalemia (serum $K^+ < 2.5$ mEq/L) can be associated with diffuse muscle weakness (3), but in most cases, hypokalemia is asymptomatic. Abnormalities in the ECG are the major manifestation of hypokalemia, and can be present in 50% of cases (13). The ECG abnormalities include prominent U waves (more than 1 mm in height), flattening and inversion of T waves, and prolongation of the QT interval. However, these changes are not specific for hypokalemia; i.e., the T wave changes and U waves can be seen with left ventricular hypertrophy, and QT prolongation has a multitude of potential offenders.

Arrhythmias

Contrary to popular belief, *hypokalemia alone is not a risk for serious arrhythmias* (3). However, hypokalemia can add to the risk of serious arrhythmias from other conditions (e.g., myocardial ischemia) (3).

Management of Hypokalemia

The first concern in hypokalemia is to eliminate or treat any condition that promotes transcellular potassium shifts (e.g., alkalosis) (3). If the hypokalemia is due to K^+ depletion, proceed as described next.

Estimate Potassium Deficits

About 10% of total body K^+ is lost for every 1 mEq/L decrease in serum K^+ (13). For a 70 kg adult with a normal total body K^+ of 50 mEq/kg, the estimated K^+ deficits associated with progressive hypokalemia are shown in Table 36.1. Note that even mild hypokalemia (serum K^+ = 3 mEq/L) is associated with a considerable K^+ deficit (175 mEq).

TABLE 36.1 Estimated Potassium Deficits in Hypokalemia†

Serum K^+ (mEq/L)	K^+ Deficit (mEq)	Serum K^+ (mEq/L)	K^+ Deficit (mEq)
3.4	35	2.9	210
3.3	70	2.8	245
3.2	105	2.7	280
3.1	140	2.6	315
3.0	175	2.5	350

†Estimates based on a lean body weight of 70 kg and a total body K^+ of 50 mEq/kg.

Potassium Replacement

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FLUIDS: The usual replacement fluid is potassium chloride, which is available as a concentrated solution (1–2 mEq/mL) in ampules containing 10, 20, 30, and 40 mEq of potassium. These solutions are *extremely hyperosmotic* (the 2 mEq/mL solution has an osmolarity of 4,000 mosm/L) and *must be diluted* (14). A potassium phosphate solution is also available that contains 4.5 mEq potassium and 3 mmol phosphate per mL, and is preferred by some for potassium replacement in diabetic ketoacidosis (because of the phosphate depletion that accompanies ketoacidosis).

RATE OF REPLACEMENT: The standard method of intravenous K^+ replacement is to add 20 mEq of K^+ to 100 mL of isotonic saline and infuse over 1 hour (15). The *maximum rate* of intravenous potassium replacement is usually set at *20 mEq/hr* (15), but dose rates of 40 mEq/hr may be necessary for severe hypokalemia, and *dose rates as high as 100 mEq/hr have been used safely* (16). Infusion through a large, central vein is preferred, if possible, because of the irritating properties of the hyperosmotic KCL solutions. However, delivery into the superior vena cava is not recommended if the desired rate of replacement exceeds 20 mEq/hr because there is a (poorly documented) risk of producing asystole.

RESPONSE: The serum K^+ may be slow to correct initially, as predicted by the flat portion of the curve in Figure 36.2. If the hypokalemia is resistant or refractory to K^+ replacement, magnesium depletion should be considered. Magnesium depletion promotes urinary K^+ loss (as described earlier), and *in patients who are magnesium deficient, hypokalemia is often resistant to K^+ replacement until the magnesium is repleted* (17). Magnesium deficiency may play an important role in diuretic-induced hypokalemia, as described in the next chapter.

HYPERKALEMIA

While hypokalemia is often well tolerated, hyperkalemia (serum $K^+ > 5.5$ mEq/L) can be a life threatening condition

Mechanisms

Hyperkalemia can be the result of potassium release from cells (transcellular shift), or impaired renal excretion of potassium. If the source of the hyperkalemia is unclear, a spot urine K^+ can be useful, as *a low urine K^+ (<30 mEq/L) is evidence of impaired renal excretion* (18). If the hyperkalemia is unexpected, the condition described next should be considered.

Factitious Hyperkalemia

Hyperkalemia can be present *ex vivo* (i.e., in the blood sample), but not *in vivo* (i.e., in the bloodstream) when K^+ is released from blood cells that are damaged when the blood specimen is obtained. This *factitious hyperkalemia* (also called "pseudohyperkalemia") usually originates from hemolysis of red blood cells, but platelets can contribute in the setting of thrombocytosis (platelet count $>500 \times 10^9$/L), and severe leukocytosis can be a factor in patients with lymphoproliferative disorders (19).

The following factors have been identified as contributing to factitious hyperkalemia (19): (*a*) the use of fist-clenching to help identify veins (which can release K^+ from skeletal muscle)

(20), (*b*) the use of excessive suction (e.g., with "vacutainers") to withdraw the blood specimen, (*c*) trauma to the specimen during transport in pneumatic tube systems (21), and (*d*) delays in processing the blood specimen.

Factitious hyperkalemia should be suspected when there is no apparent reason for hyperkalemia (the lab is often unable to identify these cases). In this situation, you should draw the blood specimen yourself, with attention to minimizing or avoiding potential contributing factors.

Transcellular Shift:

The conditions associated with K^+ movement out of cells include acidosis, rhabdomyolysis, tumor lysis syndrome, drugs, and blood transfusions.

ACIDOSIS: The presumed mechanism for the relationship between acidosis and hyperkalemia is competition between H^+ and K^+ for the same site on the membrane pump that moves K^+ into cells. However, the causal link between acidosis and hyperkalemia is being questioned because lactic acidosis is not associated with hyperkalemia (9), and respiratory acidosis has an inconsistent association with hyperkalemia (9).

TUMOR LYSIS SYNDROME: Tumor lysis syndrome is an acute, life-threatening condition that appears within 7 days after the initiation of cytotoxic therapy for selected malignancies (e.g., non-Hodgkin lymphoma). Characteristic features include the combination of hyperkalemia, hyperphosphatemia, hypocalcemia, and hyperuricemia, often accompanied by acute kidney injury (22). Hyperkalemia is the most immediate threat to life.

DRUGS: The drugs that promote K^+ movement out of cells are listed in Table 36.2. Digitalis inhibits the membrane Na^+-K^+ exchange pump, but hyperkalemia occurs only with acute (not chronic) digitalis toxicity (23). *Succinylcholine* is an ultra short-acting neuromuscular blocking agent that inhibits the membrane Na^+–K^+ exchange pump (a depolarizing effect). This normally results in a minor increase in potassium (<1 mEq/L) that lasts 5–10 minutes (24), but life-threatening increases in serum K^+ have been reported when succinylcholine is used in patients with spinal cord injury; an effect attributed to "denervation hypersensitivity" to depolarizing agents.

TABLE 36.2 Drugs that Promote Hyperkalemia

Promote Transcellular Shift	Reduce Renal Excretion
β-Blockers	ACE inhibitors
Digitalis	ARBs
Succinylcholine	Amiloride
	NSAIDs
	Spironolactone
	TMP-SMX

ARBs = angiotensive receptor blockers, TMP-SMX = trimethoprim sulfamethoxazole.

Impaired Renal Excretion

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As mentioned earlier, the capacity for urinary K^+ excretion is great enough to prevent sustained hyperkalemia after a potassium load (1). As a result, *hyperkalemia always involves (at least in part) a defect in renal potassium excretion*. This can be the result of renal disease, adrenal insufficiency, or drugs. Adrenal insufficiency impairs renal potassium excretion, but *hyperkalemia is seen only in chronic adrenal insufficiency*.

DRUGS: Drugs that impair renal potassium excretion are a common cause of hyperkalemia, and the frequent offenders are listed in Table 36.2 (24–28). All of these agents promote hyperkalemia by inhibiting some part of the renin–angiotensin–aldosterone system. (The exception is trimethoprim-sulfamethoxazole, which directly inhibits potassium secretion in the distal tubules.) Drug-induced hyperkalemia often occurs in combination with K^+ supplements or renal insufficiency.

Blood Transfusion

Hyperkalemia is a recognized (but inconsistent) complication of massive blood transfusion (i.e., blood replacement equivalent to the blood volume). The temperature used to store red blood cells (4º C) shuts off the Na^+-K^+ exchange pump in the erythrocyte cell membrane, and this results in a steady leakage of K^+ from the cells (29). The K^+ concentration in the supernatant increases steadily as the storage time increases. After 18 days of storage (the average time that blood is stored) the potassium load in one unit of packed red blood cells is 2–3 mEq (29), so massive transfusion (usually at least 6 units of red blood cells) represents a K^+ load of at least 12–18 mEq. This is a considerable load, considering that plasma contains about 9–10 mEq of K^+ in an average-sized adult.

The potassium load in transfused blood is normally cleared by the kidneys, but when systemic blood flow is compromised (which applies to most patients who need massive blood transfusions), renal K^+ excretion is impaired, and the K^+ in blood transfusions will accumulate. The transfusion volume needed to produce hyperkalemia will vary, but one study has shown that hyperkalemia begins to appear after transfusion of 7 units of red blood cells (30).

Cautopyreiophagia

In 1985, a case report was published describing a patient with severe hyperkalemia who was discovered to be ingesting 1,500 burnt match heads daily (a pica) (31). It turns out that burnt match heads are rich in potassium chlorate, and the habit of eating burnt match heads is known as *cautopyreiophagia*. This case is mentioned to demonstrate that anything is possible in clinical practice, and that there is a name for everything.

Clinical Consequences

The principal threat with hyperkalemia is slowed impulse transmission in the heart (from depolarization of cardiac muscle), which can progress to heart block and cardiac arrest.

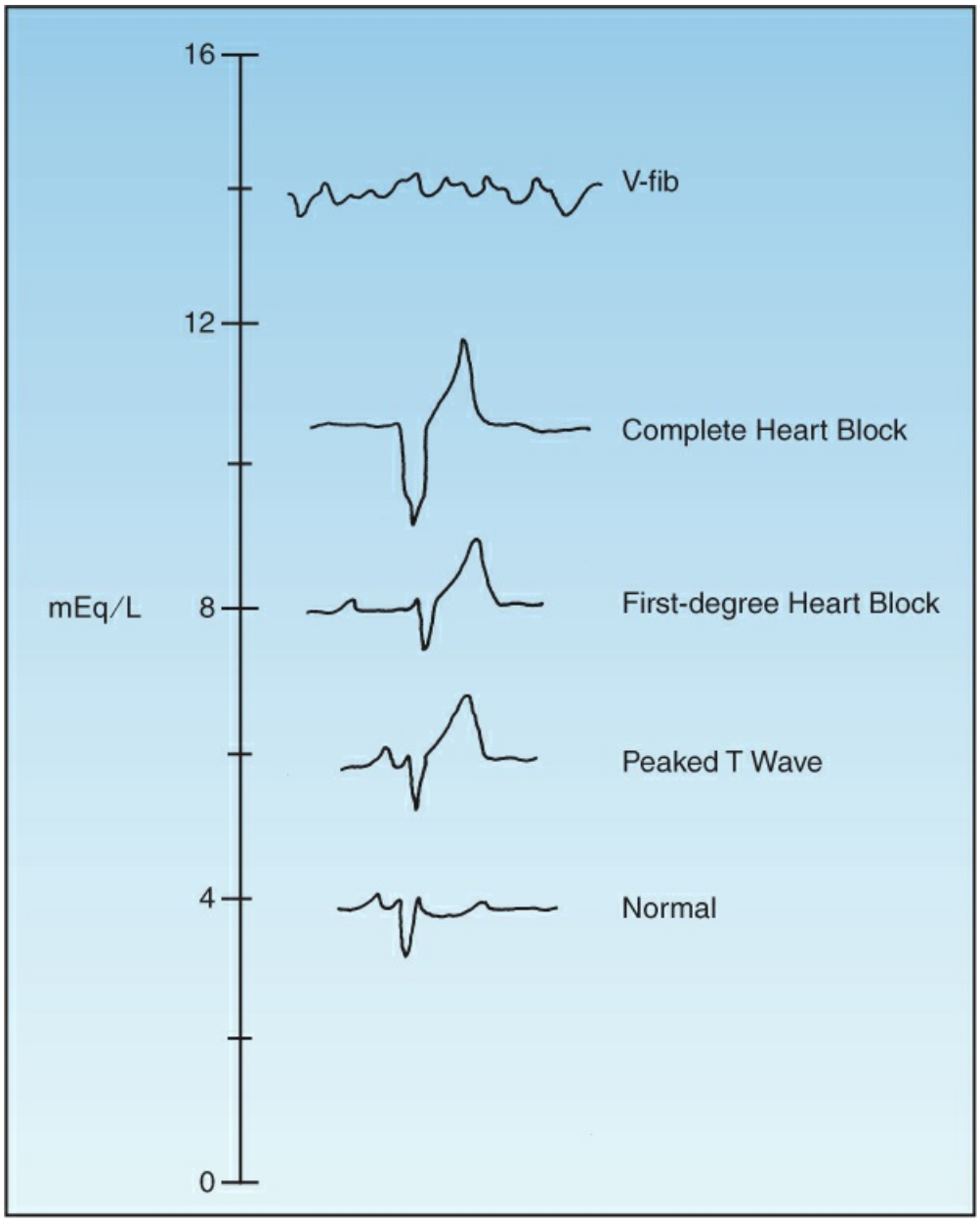


FIGURE 36.4 ECG abnormalities in progressive hyperkalemia.

ECG Changes

The expected ECG changes in progressive hyperkalemia are shown in Figure 36.4. The earliest change is the appearance of a tall, tapering (tented) T wave that is most evident in precordial leads V_2 and V_3. As the hyperkalemia progresses, the P wave amplitude decreases and the PR interval lengthens. The P waves eventually disappear and the QRS complex widens to produce a complete heart block. The final event is ventricular fibrillation or asystole.

Contrary to popular lore, *there is no predictable relationship between the severity of hyperkalemia and the presence of ECG changes* (32). For example, the traditional teaching has been that ECG changes begin to appear when the serum K^+ approaches 6 mEq/L (32), but one clinical study found that only 40% of patients with a serum K^+ above 7 mEq/L had ECG changes

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(33). Another more extreme example of this is a case report of a patient with a serum K^+ of 14 mEq/L who had no ECG changes (34). This lack of correlation between the serum K and the ECG has led to the suggestion that hyperkalemia alone may not be responsible for the ECG changes.

Management of Severe Hyperkalemia

Severe hyperkalemia is defined as a serum $K^+ > 6.5$ mEq/L, or any serum K^+ that produces ECG changes (35). The management of this condition has 3 goals (36): (*a*) blocking the effects of hyperkalemia on the cardiac conduction system, (*b*) moving K^+ into cells, and (*c*) removing excess K^+ from the body. The methods used to achieve these goals are summarized in Table 36.3.

TABLE 36.3 Management of Severe Hyperkalemia

Sequential Goals	Treatment Regimen
1. Cardiac stabilization	• 10% calcium gluconate: 10 mL IV over 3 min, and repeat after 5 min, if necessary. • Use 10% calcium chloride only for circulatory collapse (has triple the Ca^{++} content of calcium gluconate).
2. Move K^+ into cells	• Regular insulin: 10 units as an IV bolus, plus 50 mL of 50% dextrose as an IV bolus. • Albuterol: 20 mg as inhaled aerosol.
3. Eliminate Excess K^+	• Sodium zirconium cyclosilicate: 10 g every 8 hr for 48 hr, then decrease dose.

Cardiac Stabilization

Hyperkalemia has a depolarizing effect on myocardial cell membranes, which closes sodium channels and slows electrical transmission (37). These effects are antagonized by the hyperpolarizing effects of calcium, which is the treatment of choice for the cardiotoxic effects of hyperkalemia.

REGIMEN: Calcium is available in two intravenous preparations: *10% calcium gluconate*, which contains 9 mg/mL of elemental calcium in a 10 mL ampule, and *10% calcium chloride*, which contains 27 mg/mL of elemental calcium in a 10 mL ampule. Therefore, the calcium chloride preparation delivers three times more calcium than the calcium gluconate preparation. The less concentrated gluconate preparation is used in most instances, while the calcium chloride is reserved for cases of cardiovascular collapse or cardiac arrest.

The calcium regimen is shown in Table 36.3. The calcium effect should be evident after a few minutes. However, the effect dissipates in 30–60 minutes, so the other interventions in Table 36.3 should be combined with the calcium treatment.

INDICATIONS: The published indications for calcium include a serum $K^+ > 6.5$ mEq/L (with or without ECG changes), and ECG changes (regardless of the serum K^+) (35). However, some prefer to use calcium only when there are appropriate ECG changes; a strategy that seems appropriate, given the poor correlation between the serum K^+ level and the risk of ECG changes.

DIGITALIS: Calcium has traditionally been contraindicated for managing the hyperkalemia associated with digitalis toxicity (because digitalis toxicity poisons the membrane pump, which leads to life-threatening calcium accumulation in cells). However, retrospective studies have revealed numerous instances where intravenous calcium was given to patients with digitalis toxicity, without harm (38).

Moving Potassium Into Cells

There are three methods that are used to move K^+ into cells.

INSULIN-DEXTROSE: Insulin drives K^+ into skeletal muscle cells by activating the membrane Na^+–K^+ exchange pump, and the insulin-dextrose regimen in Table 36.3 will decrease the serum K^+ by at least 0.6 mEq/L (35). However, this effect is temporary (peak effect at 30–60 min), so measures that promote K^+ elimination should be started as well.

β_2-AGONISTS: Inhaled β_2-agonists (e.g., albuterol) produce a small decrease in plasma K^+ (<0.5 mEq/L) in therapeutic doses (7). The dose needed to produce a significant (0.5–1 mEq/L) drop in serum K^+ is at least 4 times the therapeutic dose (35), and this can produce unwanted side effects (e.g., tachycardia). Despite this risk, inhaled β_2-agonists have become part of the standard management protocol for severe hyperkalemia in many emergency departments. However, these agents should never be used alone to manage severe hyperkalemia.

SODIUM BICARBONATE: Sodium bicarbonate has been used to move K^+ into cells by promoting alkalosis, but there are two reasons why bicarbonate is not advised in this setting: (*a*) short-term infusions of sodium bicarbonate (up to four hours) have no effect on serum K^+ levels in the absence of a metabolic acidosis (35) and (*b*) bicarbonate can form complexes with calcium, which is counterproductive when calcium is used to antagonize the effects of hyperkalemia.

Eliminating Excess Potassium

Excess K^+ can be removed via the bowel using cation (Na-K) exchange resins, or it can be removed directly from the bloodstream with hemodialysis.

CATION EXCHANGE RESINS: Sodium polystyrene sulfonate or SPS (Kayexalate®) is the original cation exchange resin approved for use in severe hyperkalemia. It can be given orally (as a slurry) in a dose of 15 g every 6 hours, or rectally (as a retention enema) in a dose of 30–50 g every 6 hours. Although once the standard of care for severe hyperkalemia, the use of this agent is questioned because of uncertain efficacy, slow onset of action, and reports of necrotic lesions in the bowel linked to its use (39).

Another cation exchange resin, *sodium zirconium cyclosilicate* or SZC (Lokelma®) has been effective in managing hyperkalemia in outpatients (40) and recently has been shown to be equivalent to SPS for acute management of hyperkalemia (41). The SCZ resin does not have the adverse risks associated with SPS, and seems to be the favored choice at the present time. The dose is 10 grams orally, given 3 times daily for 48 hours (then the dose is reduced as needed). A clinically significant effect is reported after one hour (40).

HEMODIALYSIS: The most effective method of potassium removal is hemodialysis, which can

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produce a 1 mEq/L drop in serum K^+ after one hour, and a 2 mEq/L drop after 3 hours (35).

A FINAL WORD

Tip of the Iceberg

Considering that only 2% of the potassium in the body is outside cells, and that only 20% of the extracellular potassium is in plasma, monitoring the plasma K^+ gives us information about <1% of the potassium in the body. That might explain why hypokalemia is often clinically silent, and why there is a poor correlation between the severity of hyperkalemia and the presence of ECG changes.

That said, there are a few points in this chapter that deserve emphasis:

. Hypokalemia is not a risk for serious arrhythmias unless combined with a drug or clinical condition that can trigger arrhythmias (3).

. When hypokalemia is refractory to K^+ replacement, Mg^{++} deficiency is likely the source of the problem (17).

. When using calcium to antagonize the cardiac effects of hyperkalemia, make sure you know which preparation you are using (calcium gluconate or calcium chloride), since calcium chloride has three times more elemental calcium than calcium gluconate.

References

a. Washington HA (ed). *The Writings of Thomas Jefferson: Correspondence*, 1824.

Basics

1. Rose BD, Post TW. Potassium homeostasis. In: *Clinical Physiology of Acid-Base and Electrolyte Disorders.* 5th ed. New York, NY: McGraw-Hill, 2001; 372–402.
2. Alfonzo AVM, Isles C, Geddes C, Deighan C. Potassium disorders—clinical spectrum and emergency management. Resusc 2006; 70:10–25.
3. Schaefer TJ, Wolford RW. Disorders of potassium. Emerg Med Clin North Am 2005; 23:723–747.
4. Brown RS. Extrarenal potassium homeostasis. Kidney Int 1986; 30:116–127.
5. Sterns RH, Cox M, Feig PU, et al. Internal potassium balance and the control of the plasma potassium concentration. Medicine 1981; 60:339–354.

Hypokalemia

6. Glover P. Hypokalemia. Crit Care Resusc 1999; 1:239–251.
7. Allon M, Copkney C. Albuterol and insulin for treatment of hyperkalemia in hemodialysis patients. Kidney Int 1990; 38:869–872.
8. Lipworth BJ, McDevitt DG, Struthers AD. Prior treatment with diuretic augments the hypokalemic and electrocardiographic effects of inhaled albuterol. Am J Med 1989; 86:653–657.
9. Adrogue HJ, Madias NE. Changes in plasma potassium concentration during acute acid-base disturbances. Am J Med 1981; 71:456–467.
10. Bernard SA, Buist M. Induced hypothermia in critical care medicine: a review. Crit Care Med 2003; 31:2041–2051.
11. Salem M, Munoz R, Chernow B. Hypomagnesemia in critical illness. A common and clinically important problem. Crit Care Clin 1991; 7:225–252.
12. Gennari FJ, Weise WJ. Acid-base disturbances in gastrointestinal disease. Clin J Am Soc Nephrol 2008; 3:1861–1868.
13. Stanaszek WF, Romankiewicz JA. Current approaches to management of potassium deficiency. Drug Intell Clin Pharm 1985; 19:176–184.
14. Trissel LA. *Handbook on Injectable Drugs*. 13th ed. Bethesda, MD: Amer Soc Health System Pharmacists, 2005; 1230.
15. Kruse JA, Carlson RW. Rapid correction of hypokalemia using concentrated intravenous potassium chloride infusions. Arch

Intern Med 1990; 150:613–617.
16. Kim GH, Han JS. Therapeutic approach to hypokalemia. Nephron 2002; 92(Suppl 1):28–32.
17. Whang R, Flink EB, Dyckner T, et al. Magnesium depletion as a cause of refractory potassium repletion. Arch Intern Med 1985; 145:1686–1689.

Hyperkalemia

18. Evans KJ, Greenberg A. Hyperkalemia: a review. J Intensive Care Med 2005; 20:272–290.
19. Lábadi Á, Nagy Á, Miseta A, Kovács G. Factitious hyperkalemia in hematologic disorders. Scand J Clin Lab Invest 2016; 77:66–72.
20. Don BR, Sebastian A, Cheitlin M, et al. Pseudohyperkalemia caused by fist clenching during phlebotomy. N Engl J Med 1990; 322:1290–1292.
21. Guiheneuf R, Vuillaume L, Mangalaboyi J, et al. Pneumatic transport is critical for for leukemic patients with major leukocytosis: what precautions to measure lactate dehydrogenase, potassium, and aspartate aminotransferase? Ann Clin Biochem 2010; 47:94–96.
22. Howard SC, Jones DP, Pui C-H. The tumor lysis syndrome. N Engl J Med 2012; 364:1844–1854.
23. Krisanda TJ. Digitalis toxicity. Postgrad Med 1992; 91:273–284.
24. Ponce SP, Jennings AE, Madias N, Harington JT. Drug-induced hyperkalemia. Medicine 1985; 64:357–370.
25. Perazella MA. Drug-induced hyperkalemia: old culprits and new offenders. Am J Med 2000; 109:307–314.
26. Palmer BF. Managing hyperkalemia caused by inhibitors of the renin-angiotensin-aldosterone system. N Engl J Med 2004;351:585–592.
27. Thomas CM, Thomas J, Smeeton F, Leatherdale BA. Heparin-induced hyperkalemia. Diabetes Res Clin Pract. 2008; 80:e7–e8.
28. Perazella MA, Mahnensmith RL. Trimethoprim-sulfamethoxazole: hyperkalemia is an important complication regardless of dose. Clin Nephrol 1996; 46:187–192.
29. Vraets A, Lin Y, Callum JL. Transfusion-associated hyperkalemia. Transfus Med Rev 2011; 25:184–196.
30. Aboudara MC, Hurst FP, Abbott KC, et al. Hyperkalemia after packed red blood cell transfusion in trauma patients. J Trauma 2008; 64:S86–S91.
31. Abu-Hamden DK, Sondheimer JH, Mahajan SK. Cautopyreiophagia: cause of life-threatening hyperkalemia in a patient undergoing hemodialysis. Am J Med 1985; 79:517–519.
32. Gupta AA, Self M, Mueller M, et al. Dispelling myths and misconceptions about the treatment of acute hyperkalemia. Am J Emerg Med 2022; 52:85–91.
33. Montague BT, Ouellette JR, Buller GK. Retrospective review of the frequency of ECG changes in hyperkalemia. Clin J Am Soc Nephrol. 2008; 3:324–30.
34. Tran HA. Extreme hyperkalemia. South Med J 2005; 98:729–732.
35. Weisberg L. Management of severe hyperkalemia. Crit Care Med 2008; 36:3246–3251.
36. Rafique R, Peacock F, Armstead T, et al. Hyperkalemia management in the emergency department: An expert panel consensus. JACEP Open 2021; 2:e12572.
37. Mirvis DM, Goldberger AL. Electrocardiography. In: Libby P, Bonow RO, Mann DL, Zipes DP, eds., *Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine*. 8th ed., Philadelphia: Saunders Elsevier, 2008; 149–194.
38. Levine M, Nikkanen H, Pallin DJ. The effects of intravenous calcium in patients with digoxin toxicity. J Emerg Med 2011; 40:41–46.
39. Harel Z, Harel S, Shah PS, et al. Gastrointestinal adverse events with sodium polystyrene sulfonate (Kayexalate) use: a systematic review. Am J Med 2013; 126:264.e9–264.e24.
40. Pacjham DK, Rasmussen HS, Lavin PT, et al. Sodium zirconium cyclosilicate in hyperkalemia. N Engl J Med 2015; 372:222–231.
41. Gonzalez J, Nayyar D. Comparative efficacy of sodium zirconium cyclosilicate and sodium polystyrene sulfonate for acute hyperkalemia: a retrospective chart review. Hosp Pharm 2024; 58:159–164.

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Chapter 37

Magnesium

You can know the name of a bird in all the languages of the world, but you'll know absolutely nothing about the bird. So let's look at the bird and see what it's doing – that's what counts.

Richard Feynman (*a*)

Magnesium is essential for energy utilization in the organic world. To begin with, magnesium is the heart of the chlorophyll molecule in green plants, which captures the radiant energy in sunlight and stores it in the carbon bonds that hold carbohydrates together. Aerobic organisms then transfer this energy to adenosine triphosphate (ATP), and the release of energy from ATP requires magnesium (which is an essential cofactor for ATPase enzymes). Thus, magnesium is essential for providing us with energy from the sun, and for allowing us to utilize this energy to sustain life.

Magnesium is also essential for the proper functioning of the Na^+–K^+ exchange pump (which is a magnesium-dependent ATPase) that is responsible for the electrical gradient across cell membranes. As a result, magnesium plays an important role in the activity of excitable tissues (1–4). Magnesium also regulates the movement of calcium into smooth muscle cells, which gives it a pivotal role in the maintenance of cardiac contractile strength and peripheral vascular tone (4).

Now that's an element.

MAGNESIUM BASICS

The distribution of magnesium (Mg) in the human body is shown in Table 37.1 (5). The average-sized adult contains approximately 24 g (1 mole, or 2,000 mEq) of magnesium; a little over half is located in bone, whereas *less than 1% is located in plasma*. This lack of representation in the plasma limits the value of the plasma Mg as a measure of total body Mg (similar to the limitation described for the plasma K^+ measurement). One consequence of this is that *plasma Mg levels can be normal in the face of total body Mg depletion* (5,6).

Serum Magnesium

Serum is favored over plasma for measuring Mg levels because the anticoagulant used for plasma samples can be contaminated with citrate or other anions that bind Mg (5). The reference range for serum Mg depends on the daily Mg intake, which varies according to geographic region. The normal range of serum Mg for healthy adults in the United States is shown in Table 37.2 (7).

Converting Units

The clinical laboratory typically reports the serum Mg concentration in mg/dL (because Mg is partially bound to plasma proteins), while the medical literature typically uses mEq/L for the serum Mg concentration. The conversion is as follows:

$$\text{mEq/L} = \frac{\text{mg/dL} \times 10}{\text{mol wt}} \times \text{valence} \tag{37.1}$$

where mol wt is the molecular weight (atomic weight in the case of Mg) and valence is the number of charges on the atom or molecule. Magnesium has an atomic weight of 24 and a valence of 2, so a serum Mg concentration of 1.7 mg/dL is equivalent to (1.7 × 10)/24 × 2 = 1.4 mEq/L.

To complicate matters further, the International System of Units (SI units) uses mmol/L to express the serum Mg concentration. The conversion is: mmol/L = mEq/L × 0.5.

TABLE 37.1 Magnesium Distribution in a 70 kg Adult

Component	Mg Content (mEq)	% Total Body Mg
Bone	1,060	53
Muscle	540	27
Soft Tissue	384	19
Erythrocytes	10	0.7
Plasma	6	0.3
Total	2,000	100%

From Reference 5.

Ionized Magnesium

About 67% of the Mg in plasma is in the ionized (active) form (see Table 37.2), and the remaining 33% is either bound to plasma proteins, or chelated with divalent anions such as phosphate and sulfate (8). The standard assay for Mg measures all three fractions, so in cases of hypomagnesemia, it is not possible to determine if the problem is a decrease in the ionized (active) fraction, or a decrease in the bound fractions (e.g., hypoproteinemia) (9). The level of ionized Mg can be measured with an ion-specific electrode, but this is not routinely available.

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However, the clinical significance of monitoring ionized Mg versus total Mg is unclear.

TABLE 37.2 Reference Ranges for Serum Magnesium

	mg/dL	mEq/L	mmol/L
Total Serum Mg	1.7–2.4	1.4–2.0	0.7–1.0
Ionized Mg	—	0.8–1.1	0.4–0.6

Urinary Magnesium

Only small quantities of Mg are normally excreted in the urine (i.e., about 5–15 mEq in 24 hrs) (7). When Mg intake is deficient, the kidneys conserve Mg further, and urinary Mg excretion falls to negligible levels. This is shown in Figure 37.1 (10). Note that the plasma Mg remains in the normal range one week after starting a magnesium-free diet, while the urinary Mg excretion has dropped to negligible levels. This illustrates the relative value of urinary Mg excretion in the early detection of Mg deficiency.

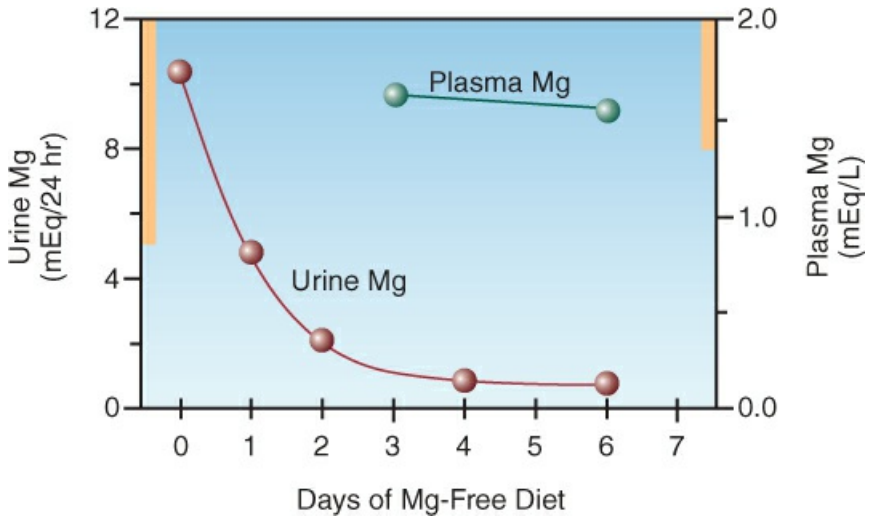


FIGURE 37.1 Urinary magnesium excretion and plasma magnesium levels in a healthy volunteer placed on a magnesium-free diet. Solid bars on the vertical axis represent the normal range for each variable. Adapted from Reference 10.

MAGNESIUM DEFICIENCY

Hypomagnesemia (total serum Mg < 1.7 mg/dL or <0.7 mmol/L) is reported in as many as 65% of ICU patients (2,11), and the frequency of magnesium deficiency is likely to be higher (because magnesium deficiency can exist without hypomagnesemia).

Predisposing Conditions

The most common predisposing conditions for magnesium depletion in the ICU are listed in Table 37.3.

TABLE 37.3 Markers of Possible Magnesium Depletion§	
Predisposing Conditions	**Clinical Findings**
Drug Therapy: Furosemide (50%) Aminoglycosides (30%) Proton Pump Inhibitors Insulin Alcohol Use Disorder (44%) Diabetes Mellitus Diarrhea (secretory) Acute MI (80%)	*Electrolyte Disorders*: Hypokalemia (40%) Hypophosphatemia (30%) Hyponatremia (27%) Hypocalcemia (22%) *Cardiac Manifestations:* Coronary Ischemia Arrhythmias Digitalis Toxicity Hyperactive CNS Syndrome

§Parentheses indicate frequency of associated hypomagnesemia.

Diuretic Therapy

Diuretics are a leading cause of Mg deficiency in ICUs. Diuretic-induced inhibition of sodium reabsorption also interferes with magnesium reabsorption, and the resultant urinary magnesium losses can parallel urinary sodium losses. Urinary magnesium excretion is most pronounced with the loop diuretics (furosemide and ethacrynic acid). *Magnesium deficiency has been reported in 50% of patients receiving chronic therapy with furosemide* (12). Thiazides have a similar tendency to produce hypomagnesemia (3). Magnesium depletion does not occur with potassium-sparing diuretics (13).

Other Drugs

The following drugs are notable for causing hypomagnesemia:

. The aminoglycosides block magnesium reabsorption in the ascending loop of Henley, and hypomagnesemia has been reported in 30% of patients receiving aminoglycosides (14).

. Proton pump inhibitors and metformin can produce hypomagnesemia by blocking magnesium absorption in the GI tract (3,15).

. Insulin and epinephrine produce hypomagnesemia by shifting magnesium into cells (3).

. Cisplatin and carboplatin can cause hypomagnesemia by promoting renal magnesium excretion (3).

Alcohol-Use Disorder

Hypomagnesemia is reported in 44% of patients with chronic alcohol-use disorder (16). Multiple factors are involved, including malnutrition, chronic diarrhea, and a blunted renal response to magnesium depletion.

Secretory Diarrhea

Secretions from the lower GI tract are rich in magnesium (10–14 mEq/L) (17), and secretory diarrhea can be accompanied by profound magnesium depletion (18). Upper GI tract secretions are not rich in magnesium (1–2 mEq/L), so vomiting does not pose a risk for magnesium depletion.

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Diabetes Mellitus

Magnesium depletion is common in insulin-dependent diabetic patients, probably as a result of urinary Mg losses that accompany glycosuria (19). Hypomagnesemia is reported in only 7% of admissions for diabetic ketoacidosis, but the incidence increases to 50% over the first 12 hours after admission (20), probably as a result of insulin-induced movement of magnesium into cells.

Acute Myocardial Infarction

Hypomagnesemia is reported in as many as 80% of patients with acute myocardial infarction (21). Several mechanisms may be involved, including an intracellular shift of Mg from excess catecholamines.

Clinical Manifestations

There are no specific clinical manifestations of Mg deficiency, but the following clinical findings can increase suspicion of Mg deficiency.

Other Electrolyte Abnormalities

Magnesium depletion is often accompanied by a decrease in serum levels of sodium, potassium, phosphate, and calcium (see Table 37.3) (22).

HYPOKALEMIA: Hypokalemia is reported in 40% of cases of Mg depletion (22). More importantly, *the hypokalemia that accompanies Mg depletion can be refractory to K^+ replacement*, and Mg replacement is necessary before the hypokalemia can be corrected (23).

HYPOCALCEMIA: Magnesium depletion can cause hypocalcemia as a result of impaired parathormone release (24) combined with an impaired end-organ response to parathormone (25). As with the hypokalemia, *the hypocalcemia that accompanies Mg depletion is difficult to correct until Mg deficits are corrected*.

HYPOPHOSPHATEMIA: Phosphate depletion is a cause rather than an effect of Mg depletion. The mechanism is enhanced renal Mg excretion (26).

Arrhythmias

Magnesium depletion will depolarize cardiac cells and promote arrhythmias. The following information regarding Mg and arrhythmias deserves mention:

. Both digitalis and Mg deficiency act to inhibit the membrane pump, so Mg deficiency will promote digitalis cardiotoxicity. Intravenous magnesium can suppress digitalis-toxic arrhythmias, even when serum Mg levels are normal (27).

. Intravenous magnesium can also abolish refractory ventricular arrhythmias when serum Mg levels are normal (28). This effect may be due to a membrane-stabilizing effect of magnesium that is unrelated to Mg repletion.

. The most cited arrhythmia linked to magnesium depletion is polymorphic ventricular tachycardia, better known as *torsade de pointes*. The role of magnesium in this arrhythmia is

presented in Chapter 19 (see Figure 19.9).

Neurologic Manifestations

The neurologic manifestations of magnesium deficiency can include altered mentation, tremors, hyperreflexia, and generalized seizures. All are uncommon and nonspecific, but the following neurotoxic syndromes deserve mention.

HYPERACTIVE CNS SYNDROME: A magnesium-related neurologic syndrome has been reported that presents with ataxia, slurred speech, excessive salivation, diffuse muscle spasms, generalized seizures, and progressive obtundation (29). The symptoms are often triggered by loud noises or bodily contact. This syndrome is associated with reduced Mg levels in cerebrospinal fluid, and it resolves with Mg infusions. The prevalence of this disorder is unknown.

WERNICKE'S: Magnesium is required for the transformation of thiamine into thiamine pyrophosphate, so *magnesium deficiency can promote thiamine deficiency, even in the face of adequate thiamine intake* (30,31).

Magnesium Replacement

The daily requirement for magnesium in healthy adults is 420 mg (35 mEq) for males and 320 mg (26 mEq) for females (see Table 37.4) (32). (*Note:* The daily requirements for magnesium are likely to be higher in ICU patients because conditions that predispose to magnesium depletion are common in these patients.) The popular oral preparation for magnesium maintenance therapy is magnesium oxide, available in 400 mg tablets, which contain 241 mg (20 mEq) of elemental magnesium (33).

Absorption of magnesium from the GI tract can be erratic in ICU patients, so intravenous magnesium is used for magnesium replacement. The standard intravenous preparation is magnesium sulfate ($MgSO_4$), which is available as magnesium sulfate heptahydrate ($MgSO_4 \cdot 7H_2O$) (34). *Each gram of this $MgSO_4$ preparation has about 8 mEq (4 mmol) of elemental magnesium* (whereas one gram of $MgSO_4$ has about twice the amount of elemental magnesium) (34). Ringer's solutions should not be used as the diluent for $MgSO_4$ because the calcium in Ringer's solutions will counteract the actions of magnesium.

TABLE 37.4 Magnesium Preparations and Daily Requirements

	Elemental Mg
Normal Daily Requirements:	
Adult Males (>30 yrs)	420 mg (35 mEq, 17.5 mmol)
Adult Females (>30 yrs)	320 mg (26 mEq, 13 mmol)
Oral Preparation:	
Magnesium Oxide (400 mg)	241 mg (20 mEq, 10 mmol)
IV Preparation:	
Magnesium Sulfate ($MgSO_4$)	98 mg (8 mEq, 4 mmol)/gram)§

The magnesium content is for magnesium sulfate heptahydrate ($MgSO_4 \cdot 7\ H_2O$), which is the usual $MgSO_4$

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preparation.

The following magnesium replacement protocols are recommended for patients with normal renal function (35). (*Note:* The $MgSO_4$ in these recommendations is for the $MgSO_4 \cdot 7H_2O$ preparation.)

Mild Hypomagnesemia

The following dosing regimen can be used for a serum Mg of 1–1.4 mEq/L (1.2– 1.7 mg/dL, or 0.5–1 mmol/L) that is clinically silent:

. Assume a total magnesium deficit of 1–2 mEq/kg.

. Because 50% of the infused magnesium can be lost in the urine, assume that the total magnesium requirement is twice the magnesium deficit.

. Replace 1 mEq/kg for the first 24 hours, and 0.5 mEq/kg daily for the next 3–5 days.

Moderate Hypomagnesemia

The following protocol is recommended for a serum Mg < 1 mEq/L (<1.2 mg/dL, or <0.5 mmol/L), or for hypomagnesemia that is accompanied by other electrolyte abnormalities:

. Add 6 g $MgSO_4$ (48 mEq of Mg) to 250 or 500 mL isotonic saline and infuse over 3 hours.

. Follow with 5 g $MgSO_4$ (40 mEq of Mg) in 250 or 500 mL isotonic saline infused over the next 6 hours.

. Continue with 5 g $MgSO_4$ every 12 hours (by continuous infusion) for the next 5 days.

Life-Threatening Hypomagnesemia

The following is recommended for hypomagnesemia associated with serious cardiac arrhythmias (e.g., torsade de pointes) or generalized seizures:

. Infuse 2 g $MgSO_4$ (16 mEq, or 8 mmol of Mg) intravenously over 2–5 minutes.

. Follow with 5 g $MgSO_4$ (40 mEq, or 20 mmol of Mg) in 250 or 500 mL isotonic saline and infuse over the next 6 hours.

. Continue with 5 g $MgSO_4$ every 12 hours (by continuous infusion) for the next 5 days.

Hypomagnesemia and Renal Insufficiency

Hypomagnesemia is not common in patients with renal insufficiency but it can occur when extrarenal Mg losses are combined with a creatinine clearance of 30–50 mL/min. When magnesium is replaced in patients with renal insufficiency, no more than 50% of the magnesium in the standard replacement protocols should be administered (35), and the serum Mg should be monitored carefully.

Monitoring Magnesium Replacement

Serum Mg levels will rise after the initial magnesium bolus, but will begin to fall after 15

minutes. Therefore, it is important to follow the bolus dose with a continuous magnesium infusion. Serum Mg levels may normalize after 1 to 2 days, but it will take several days to replenish the total body Mg stores. The magnesium retention test in Table 37.5 can be valuable for identifying the end-point of Mg replacement therapy (3,36).

TABLE 37.5 Magnesium Retention Test

Protocol:

1. Add 6 grams $MgSO_4$ (24 mmol or 48 mEq elemental Mg) to 250 mL of isotonic saline and infuse over 1 hr.
2. Collect urine for 24 hrs, beginning with the onset of the magnesium infusion.

Results:

1. Urinary Mg excretion ≤12 mmol (24 mEq) in 24 hrs (i.e., ≤50% of the infused Mg) is evidence of continued Mg depletion.
2. Urinary Mg excretion >19 mmol (38 mEq) in 24 hrs (i.e., >80% of the infused Mg) indicates sufficient Mg stores.

From Reference 36.

Magnesium Retention Test

The magnesium retention test evaluates the percentage of an infused magnesium load that is excreted in the urine. The normal rate of magnesium reabsorption is close to the maximum rate (Tmax), so most of an infused magnesium load will be excreted in the urine when magnesium stores are normal. However, when magnesium stores are deficient, the magnesium reabsorption rate is much lower than the Tmax, so more of the infused magnesium will be reabsorbed, and less will be excreted in the urine. *Magnesium deficiency is likely when less than 50% of the infused Mg is recovered in the urine, and is unlikely when more than 80% of the infused Mg is excreted in the urine* (36,37).

The magnesium retention test has one unfortunate shortcoming: i.e., it can be unreliable when renal function is impaired (which is the case in many ICU patients).

HYPERMAGNESEMIA

Hypermagnesemia is uncommon in hospitalized patients, with one survey showing a prevalence of 5% (38).

Predisposing Conditions

Most cases of hypermagnesemia are the result of impaired renal function (i.e., creatinine clearance <30 mL/min) combined with a source of magnesium intake (e.g., from magnesium-containing antacids or laxatives) (3).

Massive Hemolysis

The Mg concentration in erythrocytes is approximately three times greater than in serum (39), so hemolysis can increase the serum Mg. The serum Mg is expected to rise by 0.1 mEq/L for every 250 mL of erythrocytes that lyse completely (39), so hypermagnesemia is expected only with

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massive hemolysis.

Other Conditions

Other conditions associated with (mild) hypermagnesemia are adrenal insufficiency, hyperparathyroidism, and lithium intoxication (3).

Clinical Features

The clinical consequences of progressive hypermagnesemia are listed below (39).

Threshold Serum Mg	*Manifestation*
>4 mEq/L	Hyporeflexia
>5 mEq/L	1st° AV Block
>10 mEq/L	Complete Heart Block
>13 mEq/L	Cardiac Arrest

Magnesium has been described as *nature's physiologic calcium blocker* (40), and most of the serious consequences of hypermagnesemia are the result of calcium antagonism in the AV conduction system of the heart. Depressed contractility and vasodilation are not prominent.

Management

Hemodialysis is the treatment of choice for severe hypermagnesemia. Intravenous calcium gluconate (1 g IV over 2 to 3 minutes) can be used to antagonize the cardiovascular effects of hypermagnesemia *temporarily*, until dialysis is started (41). If fluids are permissible and some renal function is preserved, aggressive volume infusion combined with furosemide may be effective in reducing the serum magnesium levels in less advanced cases of hypermagnesemia.

A FINAL WORD

Magnesium often takes a back seat to other cations like sodium and potassium, but considering the vital roles of magnesium described in the introduction to this chapter, it deserves much more attention that it receives. The following are some specific points about magnesium that warrant emphasis:

. The serum magnesium level can be normal in patients who are magnesium depleted.

. Hypomagnesemia is reported in over 50% of ICU patients, and the frequency of magnesium depletion is likely to be even higher (because of statement #1). Diuretic therapy with furosemide is the leading cause of magnesium depletion in ICUs.

. Magnesium depletion should be suspected in any patient with diuretic-induced hypokalemia, especially when hypokalemia is refractory to potassium replacement.

. Magnesium replacement will correct the serum magnesium before total body stores of magnesium are replenished. The best indicator of magnesium repletion is the urinary retention

test (see Table 37.5).

References

a. Feynman RP. *What Do You Care What Other People Think?: Further Adventures of a Curious Character*. New York: Bantam Books, 1989.

1. Fiorentini D, Cappadone C, Farruggia G, Prata C. Magnesium: biochemistry, nutrition, detection, and social impact of diseases linked to its deficiency. Nutrients 2021; 13:1136.
2. Noronha JL, Matuschak GM. Magnesium in critical illness: metabolism, assessment, and treatment. Intensive Care Med 2002; 28:667–679.
3. Wynne Z, Falat C. Disorders of calcium and magnesium. Emerg Med Clin N Am 2023; 41:833–848.
4. Tangvoraphonkchai K, Davenport A. Magnesium and cardiovascular disease. Adv Chronic Kidney Dis 2018; 25:251–260.

Magnesium Basics

5. Elin RJ. Assessment of magnesium status. Clin Chem 1987; 33:1965–1970.
6. Reinhart RA. Magnesium metabolism. A review with special reference to the relationship between intracellular content and serum levels. Arch Intern Med 1988; 148:2415–2420.
7. Lowenstein FW, Stanton MF. Serum magnesium levels in the United States, 1971–1974. J Am Coll Nutr 1986; 5:399–414.
8. Altura BT, Altura BM. A method for distinguishing ionized, complexed and protein-bound Mg in normal and diseased subjects. Scand J Clin Lab Invest 1994; 217:83–87.
9. Kroll MH, Elin RJ. Relationships between magnesium and protein concentrations in serum. Clin Chem 1985; 31:244–246.
10. Shils ME. Experimental human magnesium deficiency. Medicine 1969; 48:61–82.

Magnesium Depletion

11. Tong GM, Rude RK. Magnesium deficiency in critical illness. J Intensive Care Med 2005; 20:3–17.
12. Dyckner T, Wester PO. Potassium/magnesium depletion in patients with cardiovascular disease. Am J Med 1987; 82:11–17.
13. Ryan MP. Diuretics and potassium/magnesium depletion. Directions for treatment. Am J Med 1987; 82:38–47.
14. Zaloga GP, Chernow B, Pock A, et al. Hypomagnesemia is a common complication of aminoglycoside therapy. Surg Gynecol Obstet 1984; 158:561–565.
15. Florentin M, Elisaf MS. Proton pump inhibitor-induced hypomagnesemia: A new challenge. World J Nephrol 2012; 1:151–154.
16. Vanoni FO, Nilani GP, Agostoni C, et al. Magnesium metabolism in chronic alcohol-use disorder: meta-analysis and systemic review. Nutrients 2021; 13:1959.
17. Kassirer J, Hricik D, Cohen J. *Repairing Body Fluids: Principles and Practice*. Philadelphia, PA: WB Saunders, 1989: 118–129.
18. Martin HE. Clinical magnesium deficiency. Ann NY Acad Sci 1969; 162:891–900.
19. Sjogren A, Floren CH, Nilsson A. Magnesium deficiency in IDDM related to level of glycosylated hemoglobin. Diabetes 1986; 35:459–463.
20. Lau K. Magnesium metabolism: normal and abnormal. In: Arieff AI, DeFronzo RA, eds. *Fluids, Electrolytes, and Acid -Base Disorders*. New York: Churchill Livingstone, 1985; 575–623.
21. Abraham AS, Rosenmann D, Kramer M, et al. Magnesium in the prevention of lethal arrhythmias in acute myocardial infarction. Arch Intern Med 1987; 147:753–755.

Clinical Manifestations

22. Whang R, Oei TO, Aikawa JK, et al. Predictors of clinical hypomagnesemia. Hypokalemia, hypophosphatemia, hyponatremia, and hypocalcemia. Arch Intern Med 1984; 144:1794–1796.
23. Whang R, Flink EB, Dyckner T, et al. Magnesium depletion as a cause of refractory potassium repletion. Arch Intern Med 1985; 145:1686–1689.
24. Anast CS, Winnacker JL, Forte LR, et al. Impaired release of parathyroid hormone in magnesium deficiency. J Clin Endocrinol Metab 1976; 42:707–717.
25. Rude RK, Oldham SB, Singer FR. Functional hypoparathyroidism and parathyroid hormone end-organ resistance in human magnesium deficiency. Clin Endocrinol 1976; 5:209–224.
26. Dominguez JH, Gray RW, Lemann J Jr. Dietary phosphate deprivation in women and men: effects on mineral and acid balances, parathyroid hormone and the metabolism of 25-OH-vitamin D. J Clin Endocrinol Metab 1976; 43:1056–1068.
27. Cohen L, Kitzes R. Magnesium sulfate and digitalis-toxic arrhythmias. JAMA 1983; 249:2808–2810.
28. Tzivoni D, Keren A. Suppression of ventricular arrhythmias by magnesium. Am J Cardiol 1990; 65:1397–1399.
29. Langley WF, Mann D. Central nervous system magnesium deficiency. Arch Intern Med 1991; 151: 593–596.
30. Dyckner T, Ek B, Nyhlin H, et al. Aggravation of thiamine deficiency by magnesium depletion. A case report. Acta Med Scand 1985; 218:129–131.

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31. Longsdale D. Thiamine and magnesium deficiencies: keys to disease. Med Hypotheses 2015; 84: 120–134.

Magnesium Replacement

32. Food and Nutrition Board, Institute of Medicine. DRI Dietary Reference Intakes for Calcium, Phosphorus, Magnesium, Vitamin D, and Fluoride. Washington, D.C.: National Academy Press, 1997.
33. Magnesium Oxide. In: *Drug Information Handbook*. 24th ed., Hudson, OH: Lexi-Comp Inc, 2015:1230.
34. Mueller M, Mitteregger T, Behringer W, et al. Be clear on magnesium (sulfate) dosing! Resuscitation 2022; 174:104.
35. Oster JR, Epstein M. Management of magnesium depletion. Am J Nephrol 1988; 8:349–354.
36. Clague JE, Edwards RH, Jackson MJ. Intravenous magnesium loading in chronic fatigue syndrome. Lancet 1992; 340:124–125.
37. Hebert P, Mehta N, Wang J, et al. Functional magnesium deficiency in critically ill patients identified using a magnesium-loading test. Crit Care Med 1997; 25:749–755.

Hypermagnesemia

38. Whang R, Ryder KW. Frequency of hypomagnesemia and hypermagnesemia. Requested vs routine. JAMA 1990; 263:3063–3064.
39. Elin RJ. Magnesium metabolism in health and disease. Dis Mon 1988; 34:161–218.
40. Iseri LT, French JH. Magnesium: nature’s physiologic calcium blocker. Am Heart J 1984; 108:188–193.
41. Mordes JP, Wacker WE. Excess magnesium. Pharmacol Rev 1977; 29:273–300.

Chapter 38

Calcium and Phosphorus

Nature never deceives us, it is always we who deceive ourselves.

Jean Jacques Rousseau (1754)

Calcium and phosphorus are responsible for much of the structural integrity of the bony skeleton. Although neither element is found in abundance in soft tissue, both have important roles in vital cell functions. Phosphorus participates in aerobic energy production, whereas calcium participates in blood coagulation, neuromuscular transmission, and smooth muscle contraction. In light of these roles, it is surprising that abnormalities in calcium and phosphorus balance are devoid of adverse consequences in most patients.

CALCIUM BASICS

Calcium is the most abundant electrolyte in the human body, (the average adult has more than half a kilogram of calcium), but 99% is in bone (1,2). Parathyroid hormone and vitamin D (calcitriol) promote calcium resorption from bone and thereby maintain plasma calcium levels, while calcitonin (from the thyroid gland) has the opposite effect and inhibits calcium release from bone.

Plasma Calcium

The calcium in plasma is present in three forms, as depicted in Figure 38.1. Approximately half of the calcium is ionized (biologically active) and the remainder is either bound to albumin (80%) or complexed with sulfate and phosphate anions (20%) (1,2). The concentration of total and ionized calcium in plasma is shown in Table 38.1. These values may vary slightly in different clinical laboratories.

TABLE 38.1 Normal Ranges for Calcium and Phosphate in Blood

Serum Electrolyte	Traditional Units (mg/dL)	Conversion Factor*	SI Units (mmol/L)
Total Calcium	9.0–10.0	0.25	2.25–2.50

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Ionized Calcium	4.6–5.0	0.25	1.15–1.25
Phosphorus	2.5–5.0	0.32	0.8–1.6

*Multiply traditional units by conversion factor to derive SI Units or divide SI Units by conversion factor to derive traditional units.

Total vs. Ionized Calcium

The calcium assay used by most clinical laboratories measures all three fractions of calcium in plasma, and this can be misleading in patients with hypoalbuminemia. This is demonstrated in Figure 38.1, which shows that a decrease in plasma albumin will decrease the total calcium concentration without affecting the ionized calcium concentration. Since the ionized calcium is the physiologically active fraction, the hypocalcemia caused by hypoalbuminemia is not physiologically significant.

A variety of correction factors have been proposed for adjusting the plasma calcium concentration in patients with hypoalbuminemia. However, none of these are reliable (3,4), and *the measurement of ionized calcium with ion-specific electrodes is necessary for accuracy.*

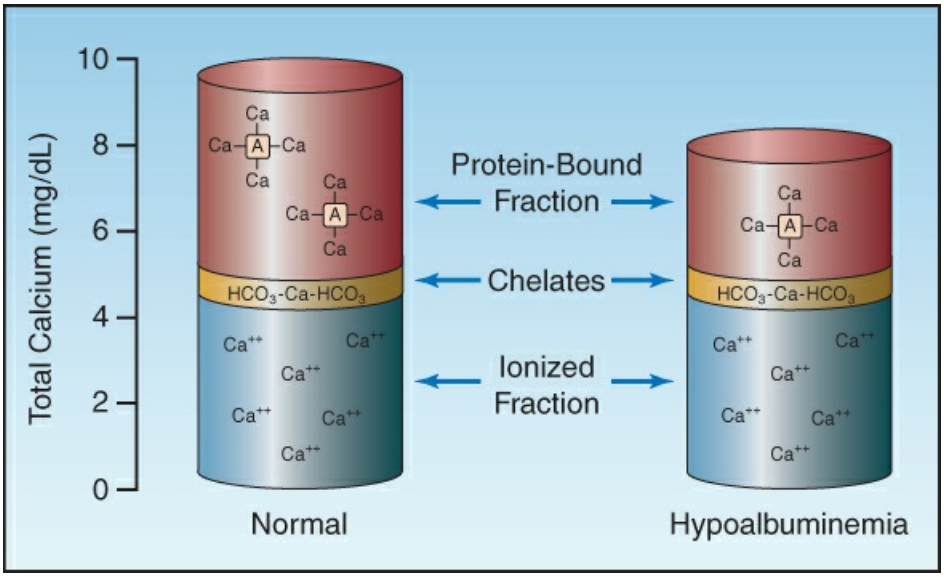


FIGURE 38.1 The three fractions of calcium in plasma and the contribution of each to the total calcium concentration. The column on the right shows how a decrease in plasma albumin can reduce the total plasma calcium without influencing the ionized (active) fraction.

Some precautions are necessary when collecting blood samples for an ionized calcium measurement. Loss of carbon dioxide from a blood sample could falsely lower the ionized calcium (by increasing calcium binding to albumin), so it is important to avoid gas bubbles in the blood sample. Anticoagulants (e.g., heparin, citrate, and EDTA) can bind calcium, so blood samples should not be placed in collection tubes that contain these anticoagulants. Tubes with red stoppers ("red top" tubes) contain silicone and are adequate for measuring ionized calcium in serum samples.

IONIZED HYPOCALCEMIA

Ionized hypocalcemia (plasma ionized Ca^{++} < 4.6 mg/dL or <1.7 mmol/L) is extremely common in ICUs (5–7), with one multicenter survey reporting an incidence of 88% in ICU patients (7). The conditions that predispose to ionized hypocalcemia in ICU patients are listed in Table 38.2.

Predisposing Conditions

Hypoparathyroidism is a leading cause of hypocalcemia in outpatients, but is not a consideration in the ICU unless a patient has had recent thyroid surgery.

TABLE 38.2 Causes of Ionized Hypocalcemia in the ICU

Most Common	Others
Alkalosis	Massive Transfusion
Magnesium Deficiency	Pancreatitis
Renal Failure	Rhabdomyolysis
Sepsis	Tumor Lysis Syndrome

Alkalosis

Alkalosis promotes the binding of calcium to albumin, and thereby reduces the fraction of ionized calcium in blood. Infusions of sodium bicarbonate can also promote ionized hypocalcemia because the infused bicarbonate forms complexes with free calcium.

Magnesium Depletion

Magnesium depletion promotes hypocalcemia by inhibiting parathormone secretion (8) and reducing end organ responsiveness to parathormone (9). *Hypocalcemia from magnesium depletion is refractory to calcium replacement therapy, and magnesium replacement often corrects the hypocalcemia*.

Systemic Inflammation

Systemic inflammation promotes ionized hypocalcemia via the actions of cytokines to stimulate calcitonin release and blunt the response to parathormone (5,6). This makes sepsis a common cause of hypocalcemia in the ICU.

Renal Failure

Renal failure can promote ionized hypocalcemia as a result of phosphate retention and impaired conversion of vitamin D to its active form in the kidneys. However, the acidosis in renal failure will decrease the binding of calcium to albumin, which helps to maintain ionized calcium levels.

Massive Transfusion

Ionized hypocalcemia is almost universal in trauma victims who receive massive blood transfusions, and the hypocalcemia can be severe (<0.9 mmol/L) in over half of the patients (10). The culprit is the citrate anticoagulant used in banked blood, which complexes with calcium.

Others

Other sources of ionized hypocalcemia include acute pancreatitis (usually severe),

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rhabdomyolysis, and tumor lysis syndrome.

Clinical Manifestations

Potential consequences of hypocalcemia include enhanced cardiac and neuromuscular excitability, and reduced contractile force in cardiac muscle and vascular smooth muscle. However, *most cases of ionized hypocalcemia in the ICU have no apparent adverse consequences* (5–7), and this has prompted the suggestion that routine monitoring of calcium should be abandoned in ICU patients (6).

Neuromuscular

The neuromuscular manifestations of hypocalcemia that are often cited include hyperreflexia, paresthesias, seizures and tetany (11). However, these rarely occur in ICU patients (5–7). *Chvostek's sign* (facial twitching when tapping the facial nerve) is cited as a sign of hypocalcemia, but this sign *has been reported in over 50% of healthy subjects with normocalcemia* (12). Trousseau's sign (i.e., carpopedal spasms in the same arm where a blood pressure cuff is inflated to exceed the systolic pressure) is also considered a reliable sign of hypocalcemia (13), but this is poorly documented.

Cardiovascular

The cardiovascular complications of hypocalcemia include hypotension, decreased cardiac output, and ventricular ectopic activity. However, these complications are rare, and are reported only in cases of extreme ionized hypocalcemia (<0.65 mmol/L) (14).

Calcium Replacement Therapy

Calcium replacement therapy is recommended only in the rare instances where hypocalcemia causes a serious adverse reaction. (The principal use of intravenous calcium is to block the cardiotoxic effects of hyperkalemia.) If calcium replacement is necessary, the intravenous calcium solutions and a recommended dosing regimen are shown in Table 38.3.

TABLE 38.3 Intravenous Calcium Replacement Therapy

Solution	Elemental Ca	Unit Volume	Osmolarity
10% Calcium chloride	27 mg/mL	10 mL ampules	2,000 mosm/L
10% Calcium gluconate	9 mg/mL	10 mL ampules	680 mosm/L

For symptomatic hypocalcemia:

1. Give a bolus dose of 200 mg elemental calcium (e.g., 22 mL of 10% calcium gluconate) in 100 mL isotonic saline over 10 minutes.
2. Follow with a continuous infusion of 1–2 mg/kg per hour for 6–12 hrs.
3. Monitor ionized calcium levels hourly for the first few hours.

Calcium Salt Solutions

The calcium solutions for intravenous use are 10% calcium chloride and 10% calcium gluconate. *Calcium chloride contains three times more elemental calcium than calcium gluconate*, but calcium gluconate is usually preferred because it has a lower osmolarity, and is less irritating

when injected.

Dosing Regimen

A bolus dose of 200 mg elemental calcium (diluted in 100 mL isotonic saline and given over 5–10 minutes) should raise the total serum calcium by 0.5 mg/dL (14), but levels will begin to fall after 30 minutes. Therefore, the bolus dose of calcium should be followed by a continuous infusion at a dose rate of 1–2 mg/kg/hr (elemental calcium) for at least 6 hours. Individual responses will vary, so calcium dosing should be guided by the level of ionized calcium in blood.

CAUTION: Intravenous calcium has been shown to promote multiorgan failure in critically ill patients (15), possibly by promoting intracellular calcium overload, which can produce a lethal cell injury (16).

Summation

Hypocalcemia in ICU patients is considered to be more of a general marker of illness than a pathological condition that requires treatment. Most cases of symptomatic hypocalcemia occur in patients with hypoparathyroidism, which is rarely encountered in the ICU.

HYPERCALCEMIA

Hypercalcemia is usually the result of a hyperparathyroidism or malignancy, with malignancy being the usual culprit in ICU patients (17). The malignancies associated with hypercalcemia include multiple myeloma, breast, lung, and ovarian cancer, squamous cell carcinoma of the head and neck, and some lymphomas (18).

Clinical Manifestations

The manifestations of hypercalcemia are varied and nonspecific. Potential manifestations can be organized as follows:

. *Gastrointestinal:* nausea, vomiting, constipation, and ileus.

. *Cardiovascular:* hypovolemia, hypotension, and shortened QT interval.

. *Renal:* polyuria and acute kidney injury.

. *Neurologic:* confusion, depressed consciousness, and coma.

Manifestations usually begin to appear when the serum calcium is >12.5 mg/dL (or the ionized calcium is >3.0 mmol/L), and they are almost always present when the serum calcium is >14 mg/dL (or the ionized calcium is >3.5 mmol/L) (17). Manifestations are more likely to appear with rapid increases in serum calcium.

Management

Treatment is indicated when the serum calcium is >14 mg/dL (or ionized calcium is >3.5 mmol/L), or when there are neurologic symptoms. The treatment options for hypercalcemia are summarized in Table 38.4

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Saline Infusion

Hypercalcemia promotes calcium loss in the urine (hypercalciuria), and this produces an osmotic diuresis that can result in hypovolemia. When this occurs, urinary calcium excretion decreases, resulting in a more rapid rise in the serum calcium. Therefore, *volume resuscitation to correct hypovolemia and promote renal calcium excretion is the first goal of management* for hypercalcemia. Isotonic saline is recommended to maintain a urine output of 100–150 mL/hr, and this can decrease the serum calcium by 1–2 mg/dL (17).

Furosemide

Loop diuretics like furosemide can increase urinary calcium excretion, but there is no evidence that intravenous furosemide adds to the benefit of saline infusions (19), and it can be counterproductive by promoting hypovolemia. Therefore, *furosemide is recommended only to alleviate volume overload* (17,19).

TABLE 38.4 Management of Severe Hypercalcemia

Agent	Dosing Regimens and Comments
Isotonic saline	Dosing: Infuse to maintain a urine output of 100–150 mL/hr. Comment: Can reduce the serum Ca^{++} by 1–2 mg/dL.
Calcitonin	Dosing: 4 Units/kg by subQ injection every 12 hrs for 24–48 hrs. Comment: Rapid onset of action (2 hrs). Tachyphylaxis is common.
Bisphosphonates	Dosing: Zoledronate (4 mg IV over 15 min) or Pamidronate (60–90 mg IV over 4 hrs). Comment: First line drugs, but have a delayed onset of action (48 hrs), and can be nephrotoxic. Zoledronate may be more effective.
Glucocorticoids	Dosing: IV Hydrocortisone (200–400 mg daily) for 3 days, then Prednisone (10–20 mg daily) for 7 days. Comment: Most effective in patients with myeloma and lymphoma.

From References 17,18.

Calcitonin

As mentioned earlier, calcitonin lowers serum calcium by inhibiting calcium release from bone. Salmon calcitonin can be given subcutaneously in a dose of 4 Units/kg every 12 hours. The response is rapid (onset within a few hours), and the serum calcium can decrease by 1–2 mg/dL (18). However, tachyphylaxis is common, and the treatment is discontinued after 24–48 hrs.

Glucocorticoids

Glucocorticoids have several effects that can decrease the serum calcium, including decreased osteoclast activity in bone, and decreased extrarenal production of calcitriol (in lymph nodes). The glucocorticoid regimen usually begins with IV hydrocortisone (200–400 mg daily) for 3 days, followed by prednisone (10–20 mg/day) for 7 days (18). Effectiveness is most evident in patients with multiple myeloma and lymphoma (17).

Bisphosphonates

The bisphosphonates are potent inhibitors of osteoclast activity. Two drugs in this class, *zoledronate* (4 mg IV over 15 min) and *pamidronate* (90 mg IV over 2 hours) are considered first line agents in the treatment of severe hypercalcemia (17,18). Zoledronate is favored because it is easier to administer, and it may be more effective than pamidronate (17,18). Both drugs have a delayed onset of action (2–4 days), and should be started as soon as the decision to treat is made. The peak effect is at 4–7 days, and the effect lasts 1–4 weeks. These agents can be nephrotoxic, and some recommend reduced dosing when the creatinine clearance is <50 mL/min (18).

Dialysis

If necessary, hemodialysis or peritoneal dialysis can be used to remove calcium in patients with renal failure.

HYPOPHOSPHATEMIA

Inorganic phosphate (PO_4) is predominantly intracellular in location, where it participates in glycolysis and high energy phosphate (ATP) production. The normal concentration of PO_4 in plasma is shown in Table 38.1 (18).

Hypophosphatemia (serum PO_4 < 2.5 mg/dL or <0.8 mmol/L) is present in as many as 80% of ICU patients (20), and is especially common after major surgery, and in multisystem trauma and fulminant sepsis.

Predisposing Conditions

Hypophosphatemia can be the result of reduced phosphate absorption from the GI tract, increased phosphate excretion in the urine, or the transcellular movement of phosphate into cells.

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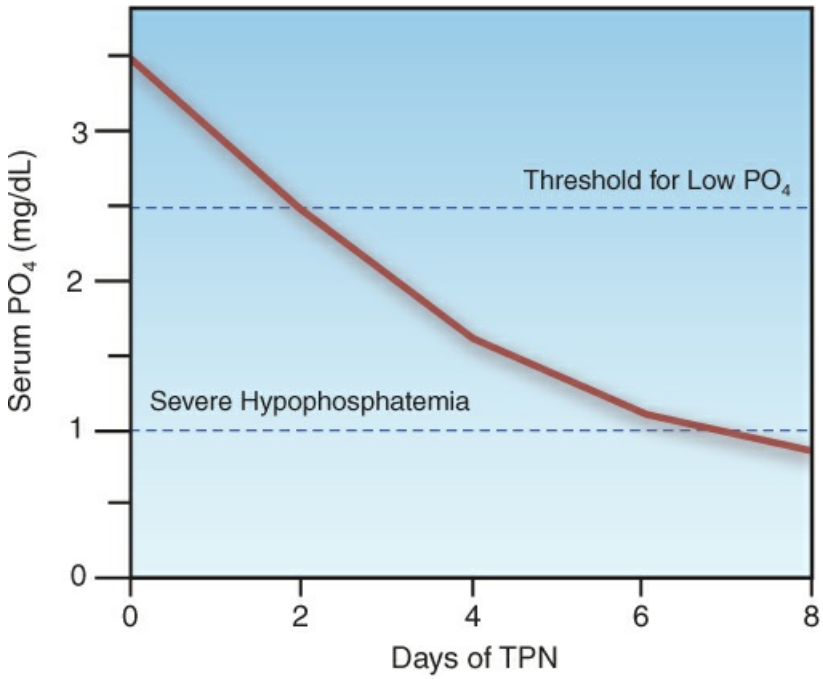


FIGURE 38.2 The effect of total parenteral nutrition (TPN) on the serum phosphate level. Data from Reference 21.

Glucose Loading

The movement of glucose into cells is accompanied by a similar movement of PO_4 into cells (presumably because PO_4 is needed for glycolysis), and this can lead to hypophosphatemia if extracellular PO_4 levels are marginal. *Glucose loading is the most common cause of hypophosphatemia in hospitalized patients* (21,22), and typically occurs with refeeding in alcoholic, malnourished, or debilitated patients. In fact, hypophosphatemia is considered a major source of *refeeding syndrome*, which can culminate in respiratory failure and cardiovascular collapse (22,23).

An example of the glucose loading effect is demonstrated in Figure 38.2, which shows the effect of total parenteral nutrition (TPN) on serum PO_4 levels (21). Note that hypophosphatemia appeared after 2 days of TPN, and reached severe levels (serum PO_4 < 1 mg/dL) after one week. TPN regimens are characteristically rich in carbohydrates (see Chapter 50), and the risk of hypophosphatemia mandates close monitoring of the serum PO_4 levels during TPN.

The glucose loading effect is also responsible for the hypophosphatemia that occurs in *diabetic ketoacidosis* (see Chapter 32). In this situation, the prolonged hyperglycemia results in an osmotic diuresis (from glycosuria) that promotes urinary PO_4 losses, and the hypophosphatemia appears when insulin is given (which drives glucose and PO_4 into cells).

Respiratory Alkalosis

Respiratory alkalosis can increase intracellular pH, and this accelerates glycolysis. The increase in glucose utilization is then accompanied by an increase in glucose and PO_4 movement into cells (24). This may be an important source of hypophosphatemia in ventilator-dependent patients, because overventilation and respiratory alkalosis are common in these patients.

Physiological Stress

Stimulation of β-adrenergic receptors can move PO_4 into cells, and this creates a risk of hypophosphatemia during periods of physiological stress (from the actions of endogenous catecholamines). This is demonstrated by the common association of hypophosphatemia with panic attacks (25).

Systemic Inflammation

There is an inverse relationship between serum PO_4 and circulating levels of inflammatory cytokines (26). Possible explanations include increased PO_4 utilization by activated neutrophils, and a transcellular shift of PO_4 caused by endogenous catecholamines.

Phosphate Binding Agents

Aluminum can form insoluble complexes with inorganic phosphates. As a result, aluminum-containing compounds such as sucralfate can impede the absorption of phosphate in the upper GI tract and promote phosphate depletion (27).

Clinical Manifestations

Hypophosphatemia is often clinically silent, even when the serum PO_4 falls to extremely low levels. In one study of patients with severe hypophosphatemia (i.e., serum $PO_4 < 1$ mg/dL), none of the patients showed evidence of harm (28). However, severe phosphate depletion creates a risk for impaired oxidative metabolism.

Oxidative Metabolism

Phosphate depletion has several effects that could impair oxidative metabolism. These are summarized below, and indicated in Figure 38.3.

. *Cardiac output:* Phosphate depletion can impair myocardial contractility and reduce cardiac output. Hypophosphatemic patients with heart failure have shown improved cardiac performance after phosphate supplementation (29).

. *Erythrocytes:* Reduction of high energy phosphate production from glycolysis in erythrocytes can reduce the deformability of red cells. This may explain why severe hypophosphatemia can be accompanied by a hemolytic anemia (30).

. *Oxyhemoglobin Dissociation:* Phosphate depletion is accompanied by depletion of 2,3 diphosphoglycerate, and this shifts the oxyhemoglobin dissociation curve to the left. When this occurs, hemoglobin is less likely to release oxygen to the tissues.

. *Energy Availability:* Phosphate depletion can reduce the availability of cellular energy by impeding the production of high-energy phosphate compounds (ATP).

Muscle Disorders

Hypophosphatemia is a reported cause of rhabdomyolysis (31) and respiratory muscle weakness (32), but the clinical significance of these associations is unclear.

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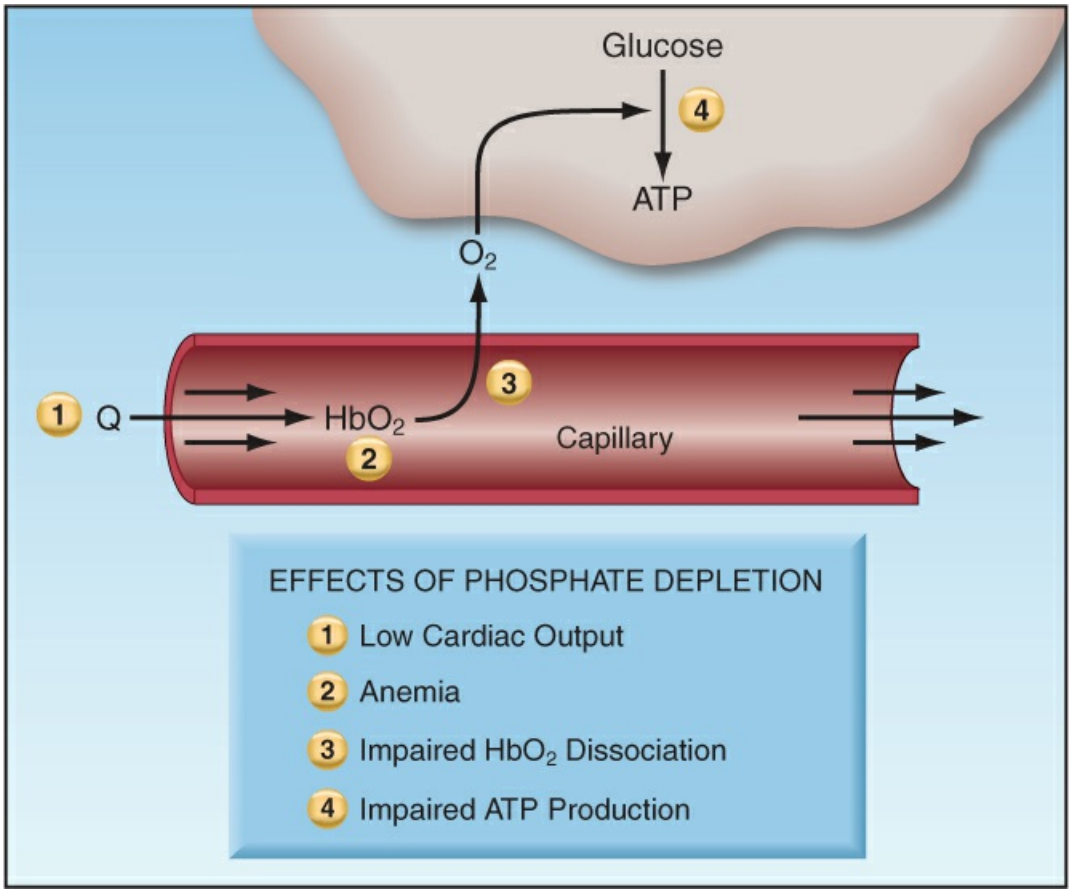


FIGURE 38.3 The effects of hypophosphatemia that threaten oxidative metabolism and cellular energy production. See text for explanation

Phosphate Replacement

Intravenous phosphate replacement is recommended for all patients with severe hypophosphatemia (i.e., serum PO_4 < 1.0 mg/dL or 0.3 mmol/L), and for patients with hypophosphatemia of any degree who also have cardiac dysfunction, respiratory failure, or rhabdomyolysis. The phosphate solutions and dosing regimen are shown in Table 38.5 (33).

TABLE 38.5 Phosphate Replacement Therapy

Solution†	PO_4 Content	Other Content
Sodium Phosphate Potassium Phosphate	93 mg (3 mmol)/mL 93 mg (3 mmol)/mL	Na^+: 4.0 mEq/L K^+: 4.3 mEq/L

PO_4 Replacement (IV) by Body Weight			
Serum PO_4 (mg/dL)	40–60 kg	61–80 kg	81–120 kg
<1	30 mmol	40 mmol	50 mmol
1–1.7	20 mmol	30 mmol	40 mmol
1.8–2.5	10 mmol	15 mmol	20 mmol

†If the plasma K^+ is ≥4 mEq/L, use sodium phosphate, and if the plasma K^+ is <4 mEq/L, use potassium phosphate.

From Reference 33

Summation

Like hypocalcemia, hypophosphatemia is common and often clinically silent, and is considered more of a general marker of illness than a pathological condition that deserves corrective action (20).

HYPERPHOSPHATEMIA

The principal cause of hyperphosphatemia in the ICU is end-stage renal disease (usually chronic, and often dialysis-dependent). Less common sources are tumor lysis syndrome and hypoparathyroidism.

Clinical Manifestations

The consequences of hyperphosphatemia include the formation of insoluble calcium–phosphate complexes (with deposition into soft tissues), and hypocalcemia (11). However, neither of these manifestations has any proven significance in ICU patients.

Management

The management of hyperphosphatemia usually involves the use of phosphate binders to reduce GI absorption of phosphate.

Phosphate Binders

Phosphate binders are separated into calcium-containing and non-calcium-containing products. Some prefer the non-calcium-containing binders because there is a daily limit of calcium intake (<800 mg) in patients with chronic kidney disease. However, there is no evidence that one type of phosphate binder is superior to the other (34). The following are examples of each type of phosphate binder.

SEVELAMER: Sevelamer is a non-calcium-containing phosphate binder that is provided as a powder, and is reconstituted in water. The dose is determined by the serum PO_4 level (35):

Serum PO_4 (mg/dL)	Dosage
5.6–7.4	800 mg TID
7.5–8.9	1,200–1,600 mg TID
>9	1,600 TID

(TID is three times daily). Sevelamer gluconate is preferred to sevelamer hydrochloride because the latter preparation can cause a metabolic acidosis (36).

CALCIUM ACETATE: Calcium acetate (PhosLo®) is available in tablets (667 mg per tab) or an oral solution (667 mg/5 mL). The initial dose is 1,334 mg (2 tabs or 10 mL) three times daily, and

this can be increased to achieve a serum PO_4 < 6 mg/dL, as long as hypercalcemia does not develop (37).

A FINAL WORD

The Normalization Heuristic

Disorders of calcium and phosphate in ICU patients are notable for the lack of apparent adverse consequences in most patients. This is especially true for hypocalcemia and hypophosphatemia, which seem to be general markers of illness rather than pathologic disorders that need corrective treatment.

The practice of correcting abnormal electrolyte values, even when there is no apparent harm, has been termed the *normalization heuristic* (38), and has two characteristic features: it is very common, and is lacking in merit. It is important to emphasize that the normal range for electrolytes is a statistically-determined domain (i.e., two standard deviations around the mean), and values outside this domain can be normal outliers, and not pathological entities that need corrective action.

References

Calcium Basics

1. Baker SB, Worthley LI. The essentials of calcium, magnesium and phosphate metabolism: part I. Physiology. Crit Care Resusc 2002; 4:301–306.
2. Wynne Z, Falat C. Disorders of calcium and magnesium. Emerg Med Clin N Am 2023; 41:833–848.
3. Smith JD, Wilson S, Schneider HG. Misclassification of calcium status based on albumin-adjusted calcium: studies in a tertiary hospital setting. Clin Chem 2018; 64:1713–1722.
4. Slomp J, van der Voort PH, Gerritsen RT, et al. Albumin-adjusted calcium is not suitable for diagnosis of hyper- and hypocalcemia in the critically ill. Crit Care Med 2003; 31:1389–1393.

Ionized Hypocalcemia

5. Melchers M, van Zanten ARH. Management of hypocalcemia in the critically ill. Curr Opin Crit Care 2023; 29:330–338.
6. Aberegg SK. Ionized calcium in the ICU. Should it be measured and corrected? Chest 2016; 149:846–855.
7. Moritoki E, Kim I, Nichol A, et al. Ionized calcium concentration and outcome in critical illness. Crit Care Med 2011; 39:314–321.
8. Anast CS, Winnacker JL, Forte LR, et al. Impaired release of parathyroid hormone in magnesium deficiency. J Clin Endocrinol Metab 1976; 42:707–717.
9. Rude RK, Oldham SB, Singer FR. Functional hypoparathyroidism and parathyroid hormone end-organ resistance in human magnesium deficiency. Clin Endocrinol 1976; 5:209–224.
10. Giancarelli A, Birrer KL, Alban RF, et al. Hypocalcemia in trauma patients receiving massive transfusion. J Surg Res. 2016; 202:182–187.
11. Baker SB, Worthley LI. The essentials of calcium, magnesium and phosphate metabolism: part II. Disorders. Crit Care Resusc 2002;4:307–315.
12. Méneret A, Guey S, Degos B. Chvostek sign, frequently found in healthy subjects, is not a useful clinical sign. Neurology 2013; 80:1067
13. van Bussel BCT, Koopmans RP. Trousseau's sign at the emergency department. BMJ Case Rep 2016; 2016.
14. Zaloga GP. Hypocalcemia in critically ill patients. Crit Care Med 1992; 20:251–262.
15. Collage RD, Howell GM, Zhang X, et al. Calcium supplementation during sepsis exacerbates organ failure and mortality via calcium/calmodulin-dependent protein kinase kinase signaling. Crit Care Med 2013; 41:e352–e360.
16. Trump BF, Berezesky IK. Calcium-mediated cell injury and cell death. Faseb J 1995; 9:219–228.

Hypercalcemia

17. Maier JD, Levine SN. Hypercalcemia in the intensive care unit: A review of pathophysiology, diagnosis, and modern therapy.

J Intensive Care Med 2015; 30:235–252.
18. Almuradova E, Cicin I. Cancer-related hypercalcemia and potential treatments. Front Endocrinol 2023; 14:1039490.
19. LeGrand S, Leskuski D, Zama I. Narrative review: furosemide for hypercalcemia: an unproven yet common practice. Ann Intern Med. 2008; 149:259–263.

Hypophosphatemia

20. Sin JCK, King L, Ballard E, et al. Hypophosphatemia and outcomes in ICU: A systematic review and meta-analysis. J Intensive Care Med 2021; 36:1025–1035.
21. Knochel JP. The pathophysiology and clinical characteristics of severe hypophosphatemia. Arch Intern Med 1977; 137:203–220.
22. Marinella MA. Refeeding syndrome and hypophosphatemia. J Intensive Care Med 2005; 20:155–159.
23. da Silva JSV, Seres DS, Sabino K, et al. ASPEN consensus recommendations for refeeding syndrome. Nutr Clin Pract 2020; 35:178–195.
24. Paleologos M, Stone E, Braude S. Persistent, progressive hypophosphataemia after voluntary hyperventilation. Clin Sci 2000; 98:619–625.
25. Roestel C, Hoeping W, Deckert J. Hypophosphatemia in panic disorder. Am J Psychiatry 2004; 161:1499–1500.
26. Barak V, Schwartz A, Kalickman I, et al. Prevalence of hypophosphatemia in sepsis and infection: the role of cytokines. Am J Med 1998; 104:40–47.
27. Miller SJ, Simpson J. Medication-nutrient interactions: hypophosphatemia associated with sucralfate in the intensive care unit. Nutr Clin Pract 1991; 6:199–201.
28. King AL, Sica DA, Miller G, et al. Severe hypophosphatemia in a general hospital population. South Med J 1987; 80:831–835.
29. Davis SV, Olichwier KK, Chakko SC. Reversible depression of myocardial performance in hypophosphatemia. Am J Med Sci 1988; 295:183–187.
30. Klock JC, Williams HE. Mentzer WC. Hemolytic anemia and somatic cell dysfunction in severe hypophosphatemia. Arch Intern Med 1974: 134:360–364.
31. Knochel JP, Barcenas C, Cotton JR, et al. Hypophosphatemia and rhabdomyolysis. J Clin Invest 1978; 62:1240–1246.
32. Gravelyn TR, Brophy N, Siegert C, et al. Hypophosphatemia-associated respiratory muscle weakness in a general inpatient population. Am J Med 1988; 84:870–876.
33. Taylor BE, Huey WY, Buchman TG, et al. Treatment of hypophosphatemia using a protocol based on patient weight and serum phosphorus level in a surgical intensive care unit. J Am Coll Surg 2004; 198:198–204.

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34. Scialla JJ, Kendrick J, Uribarri J, et al. Management of hyperphosphatemia in patients with CKD: An NKF-KDOQI controversies perspective. J Kidney Dis 2021; 77:132–141.
35. Sevelamer. In: *Drug Information Handbook*. 24th edition. Hudson, OH: Lexi-Comp Inc., 2015; 1821–1822.
36. Brezina B, Quinibi WY, Nolan CR. Acid loading during treatment with sevelamer hydrochloride: mechanisms and clinical implications. Kidney Int 2004; 66(Suppl 90):S39–S45.
37. Calcium Acetate. In: *Drug Information Handbook*. 24th edition. Hudson, OH: Lexi-Comp Inc., 2015; 318–319.

A Final Word

38. Aberegg SK, O'Brien JM. The normalization heuristic: An untested hypothesis that may misguide medical decisions. Med Hypotheses 2009; 72:745–748.

A L G r a w a n y